



POLITÉCNICA
Instituto de Energía Solar

Fundamentals of photovoltaic
energy conversion and
conventional solar cells

A.Martí

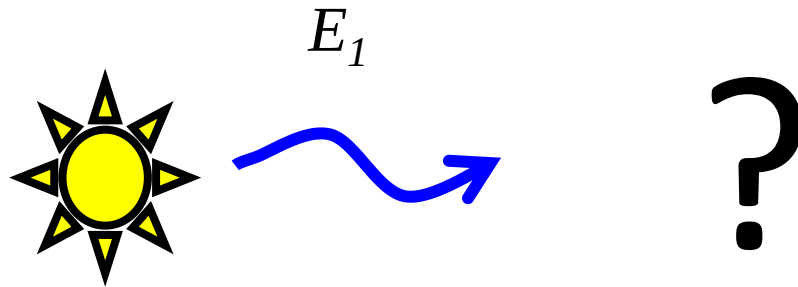
17-20 September 2018,
MATENER 2018

ICMAB, Campus UAB, Barcelona

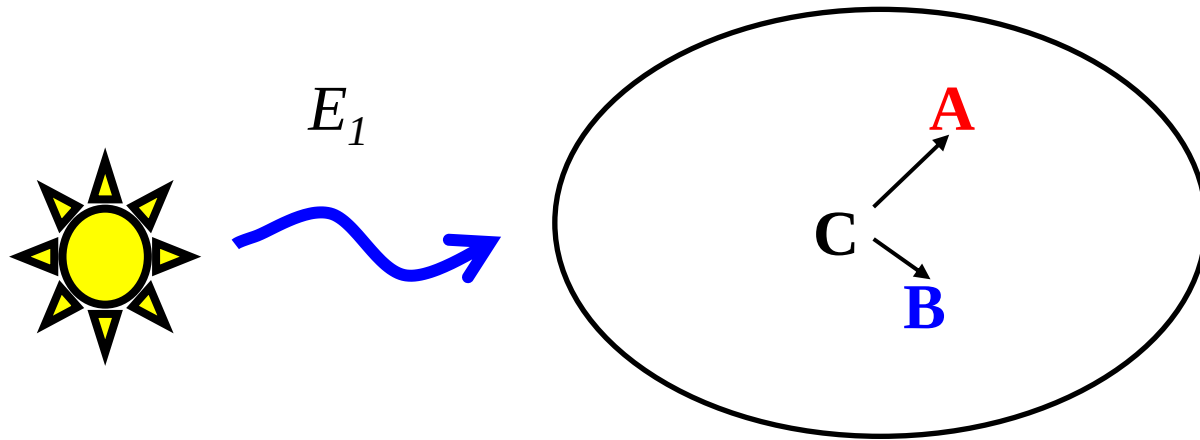
- Fundamentals of photovoltaic energy conversion
- Conventional (inorganic) solar cells

- Fundamentals of photovoltaic energy conversion
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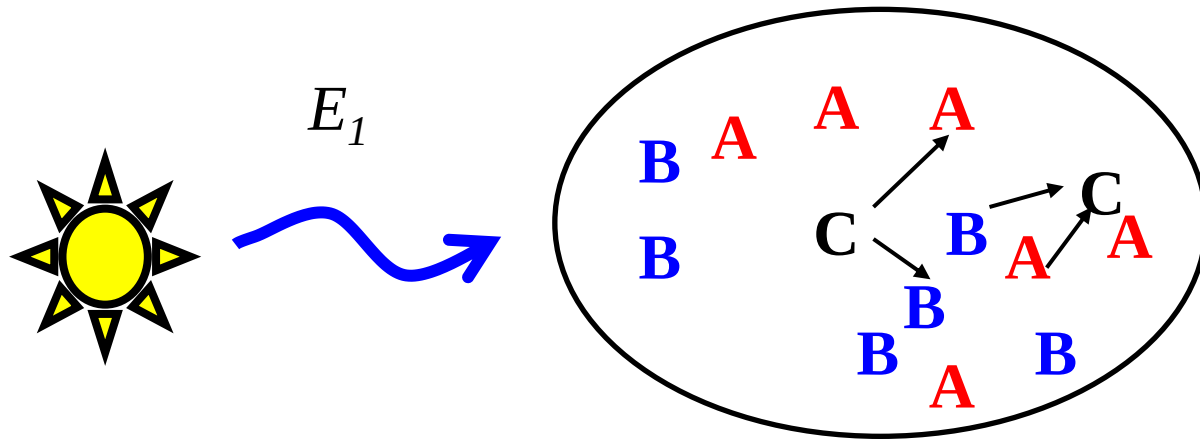
What is needed for a PV converter?



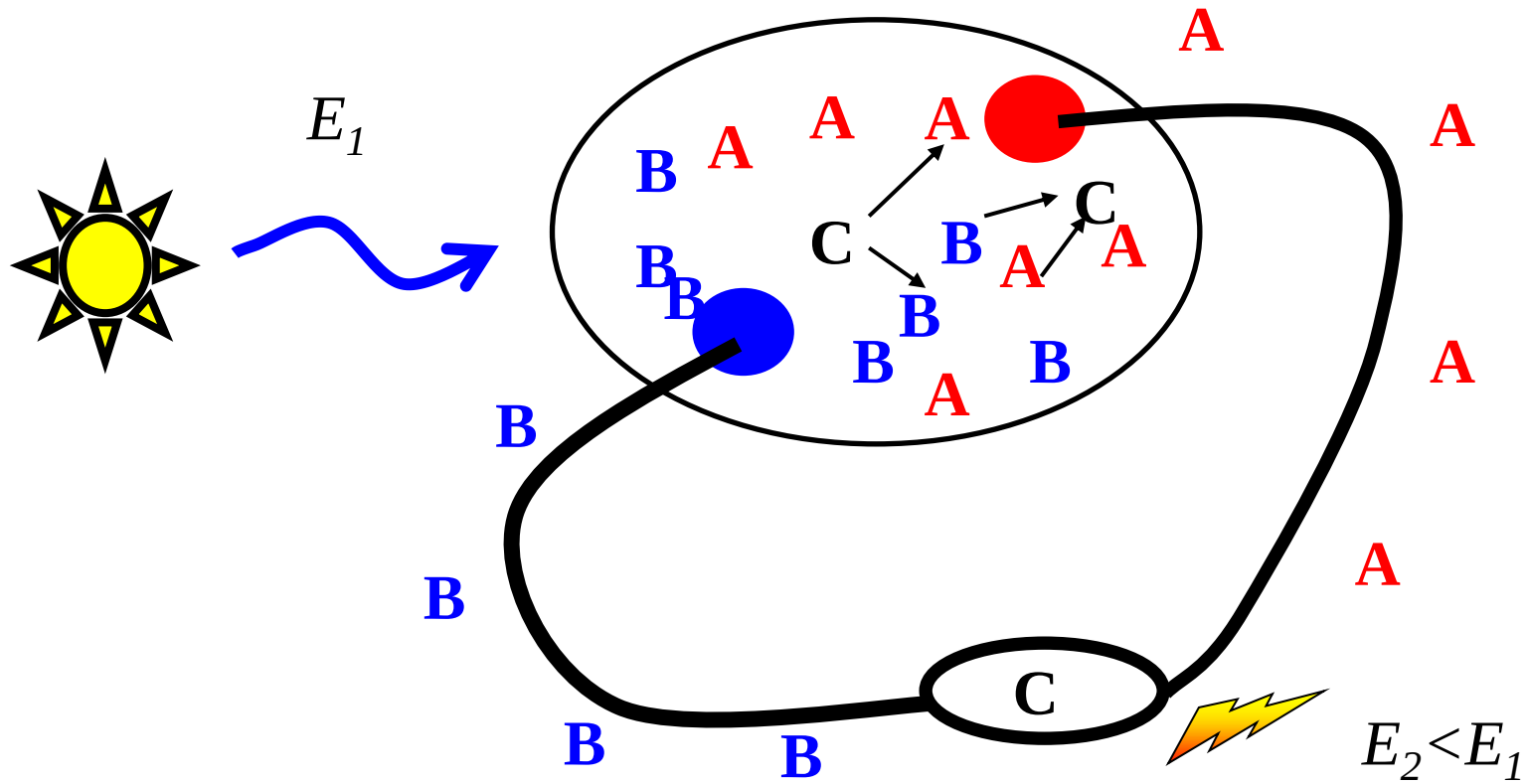
What is needed for a PV converter? A material sensitive to light



What is needed for a PV converter?: Transport x lifetime

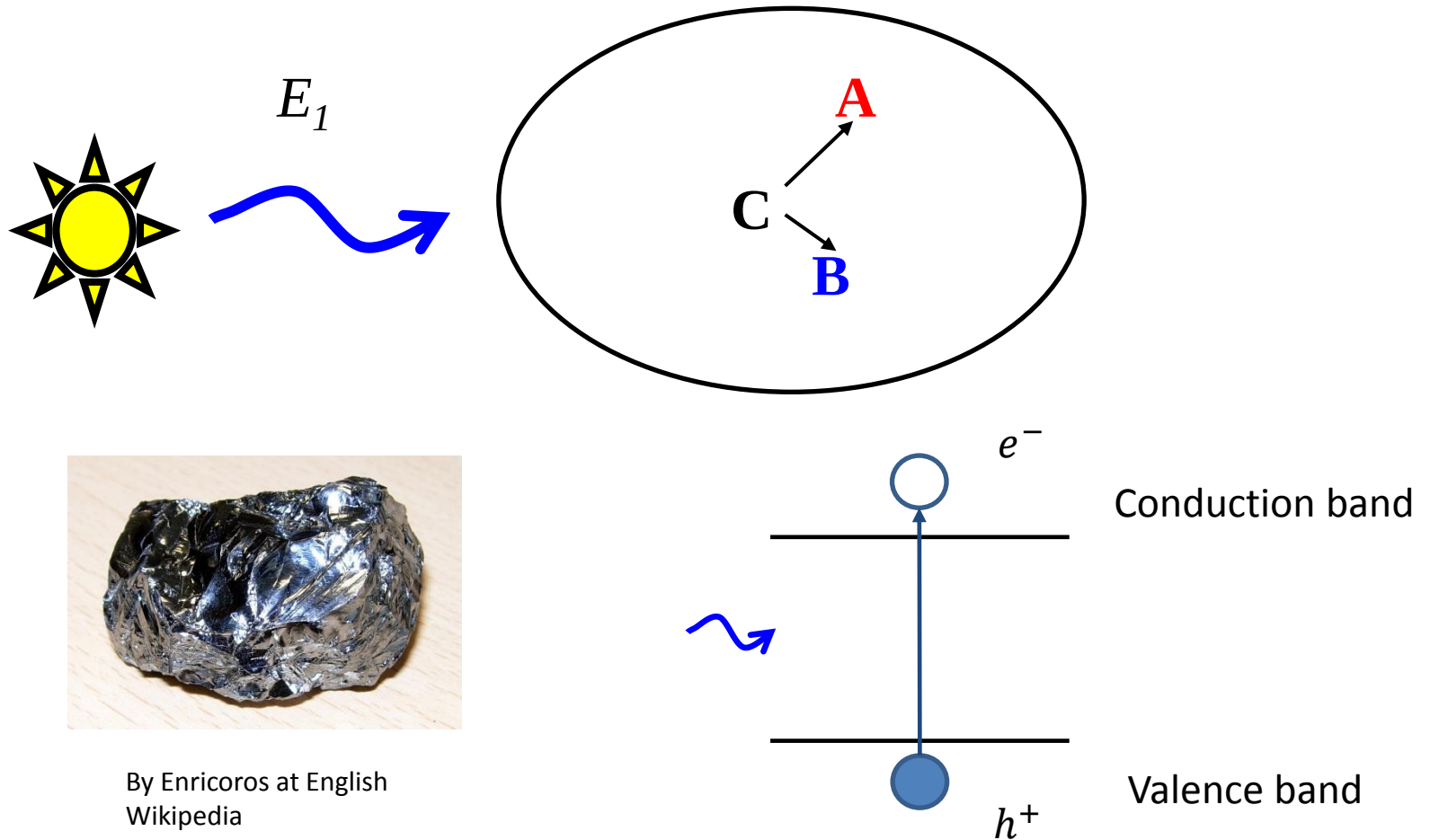


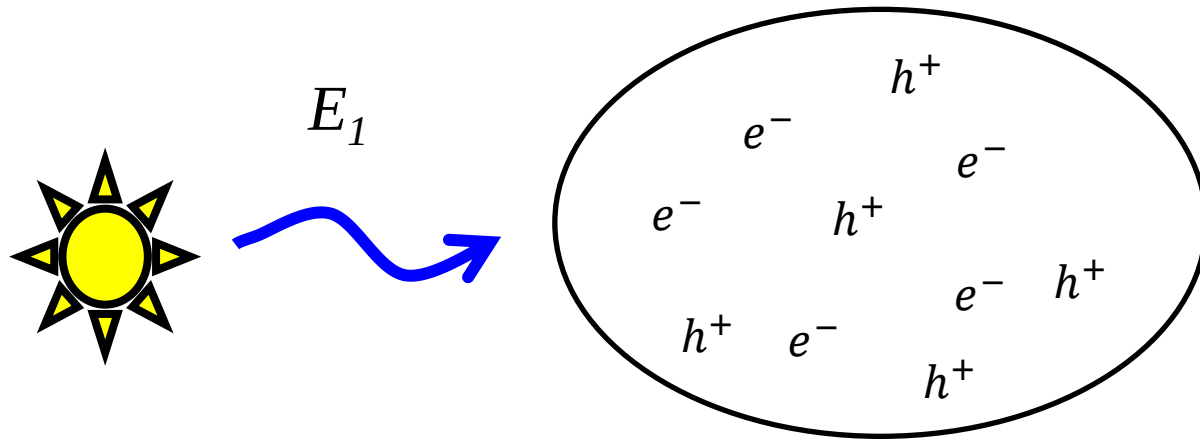
What is needed for a PV converter? Selective contacts



$$\text{Efficiency} \equiv \eta = \frac{E_2}{E_1}$$

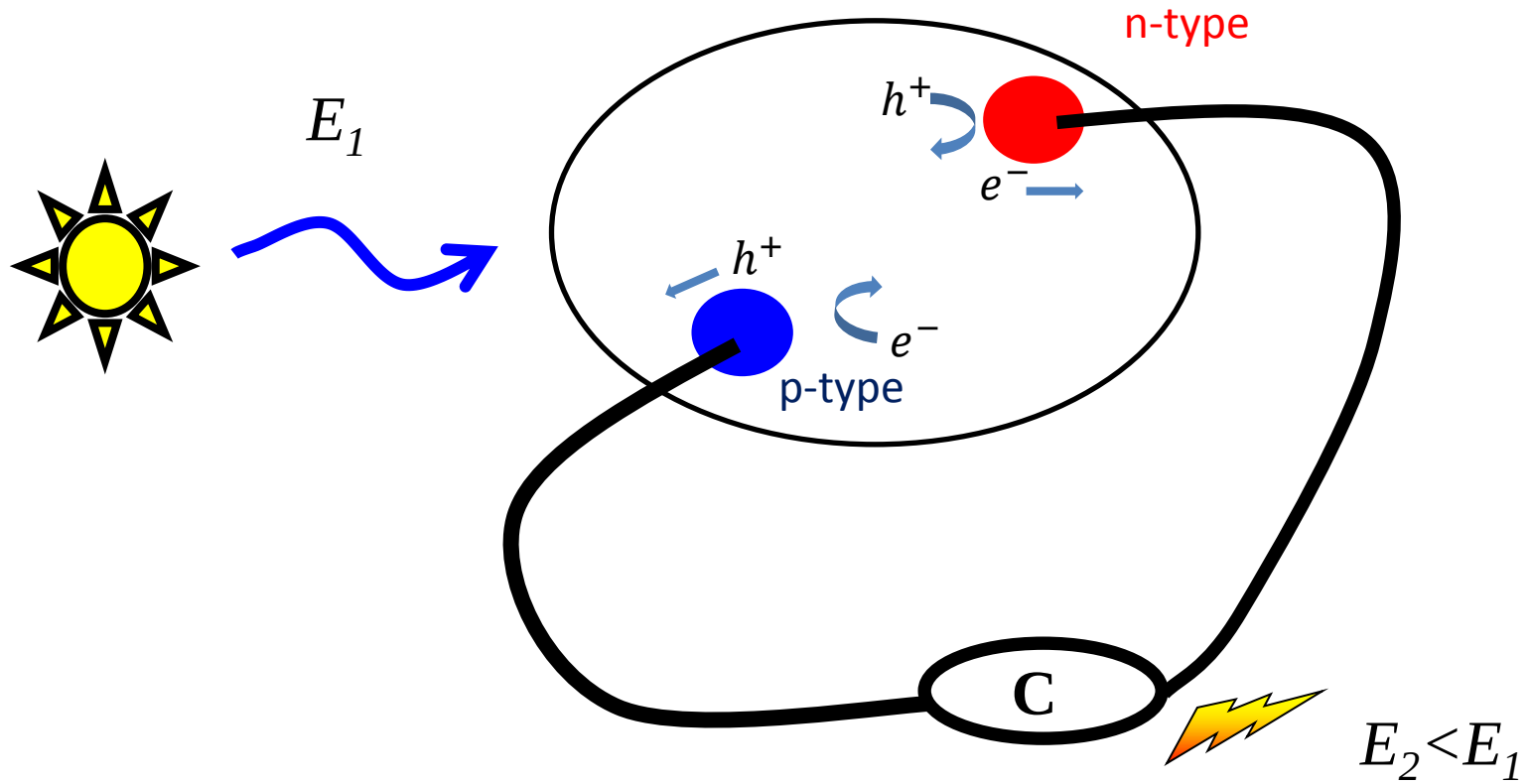
A material sensitive to light: A semiconductor



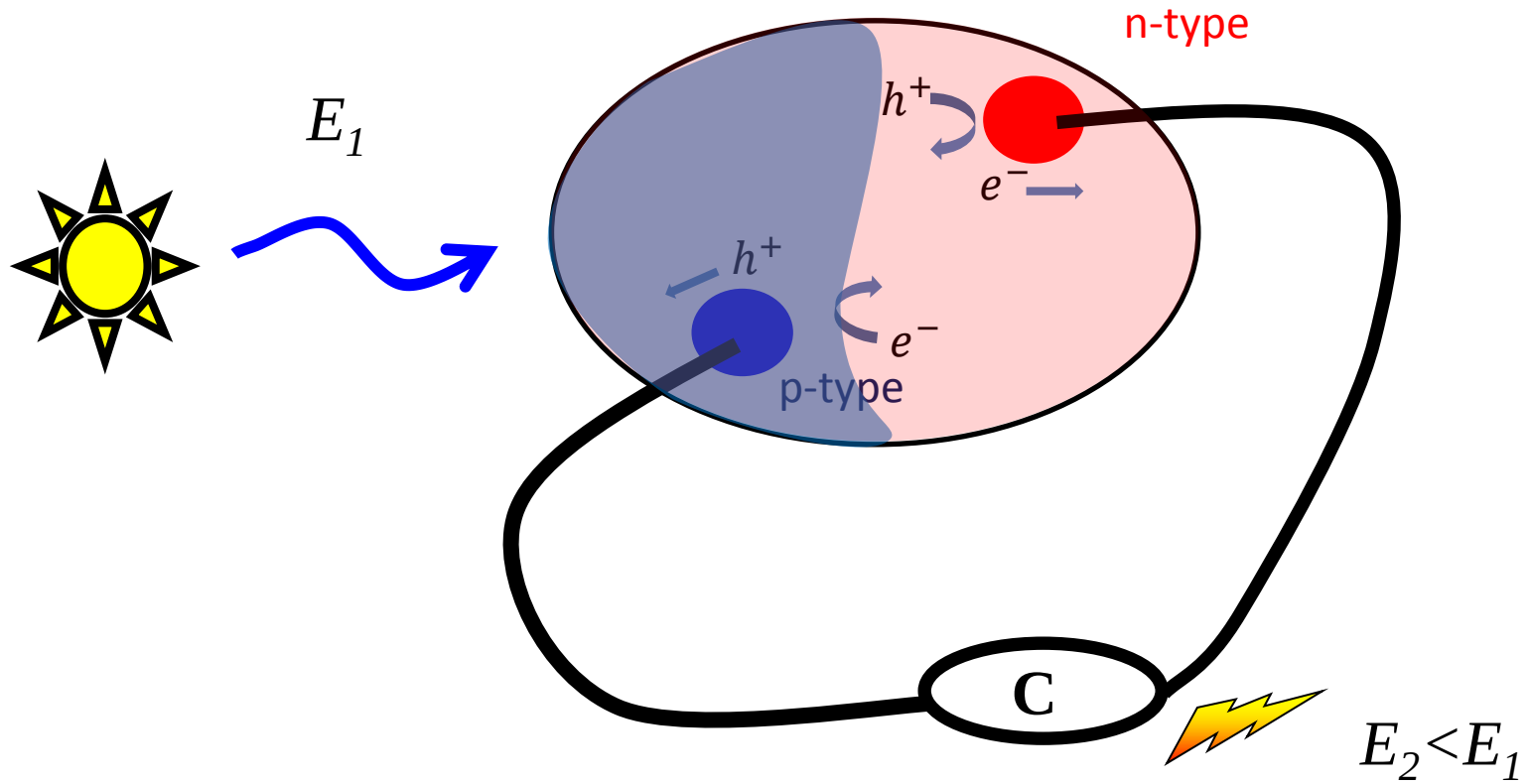


$$\text{Diffusion length} \propto \sqrt{\text{mobility} \times \text{lifetime}}$$

Selective contacts: p-type and n-type semiconductors



Selective contacts: p-type and n-type semiconductors



The solar cell as pn junction

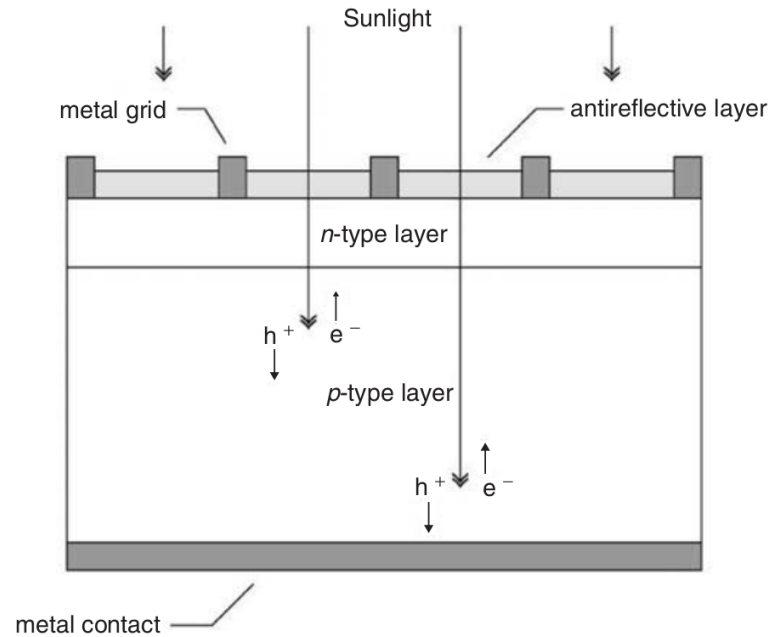
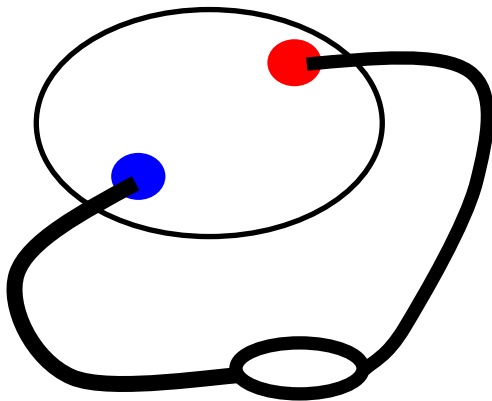
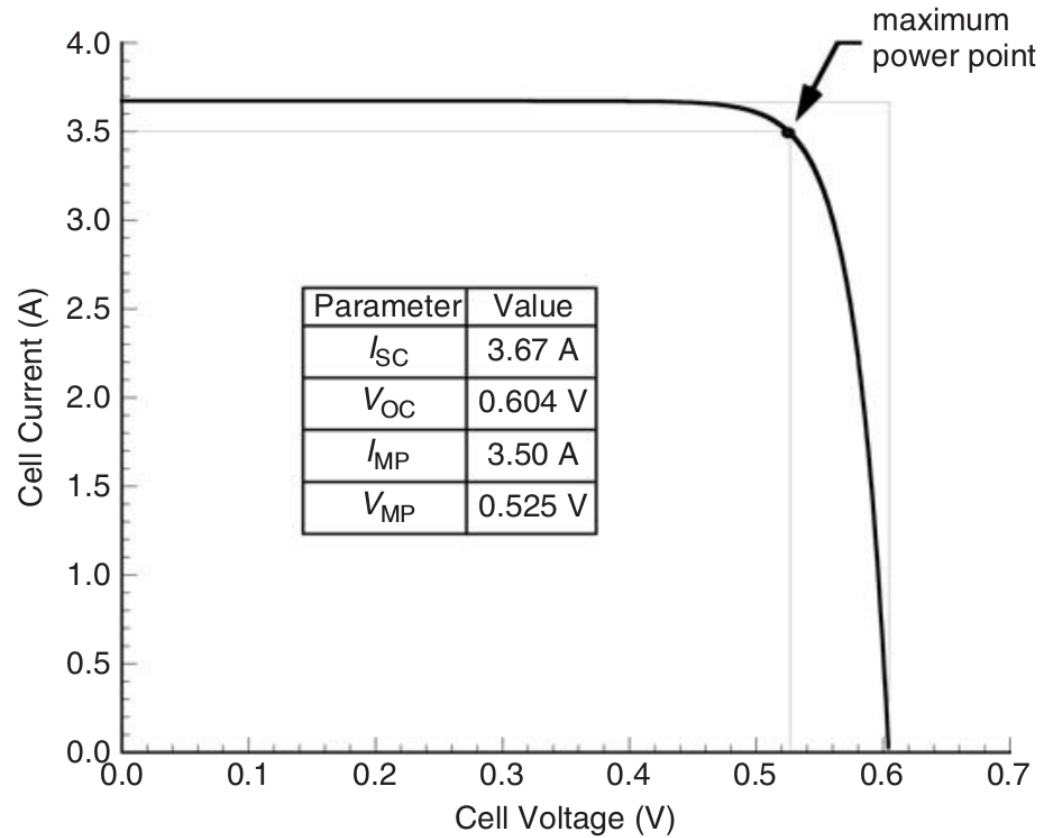
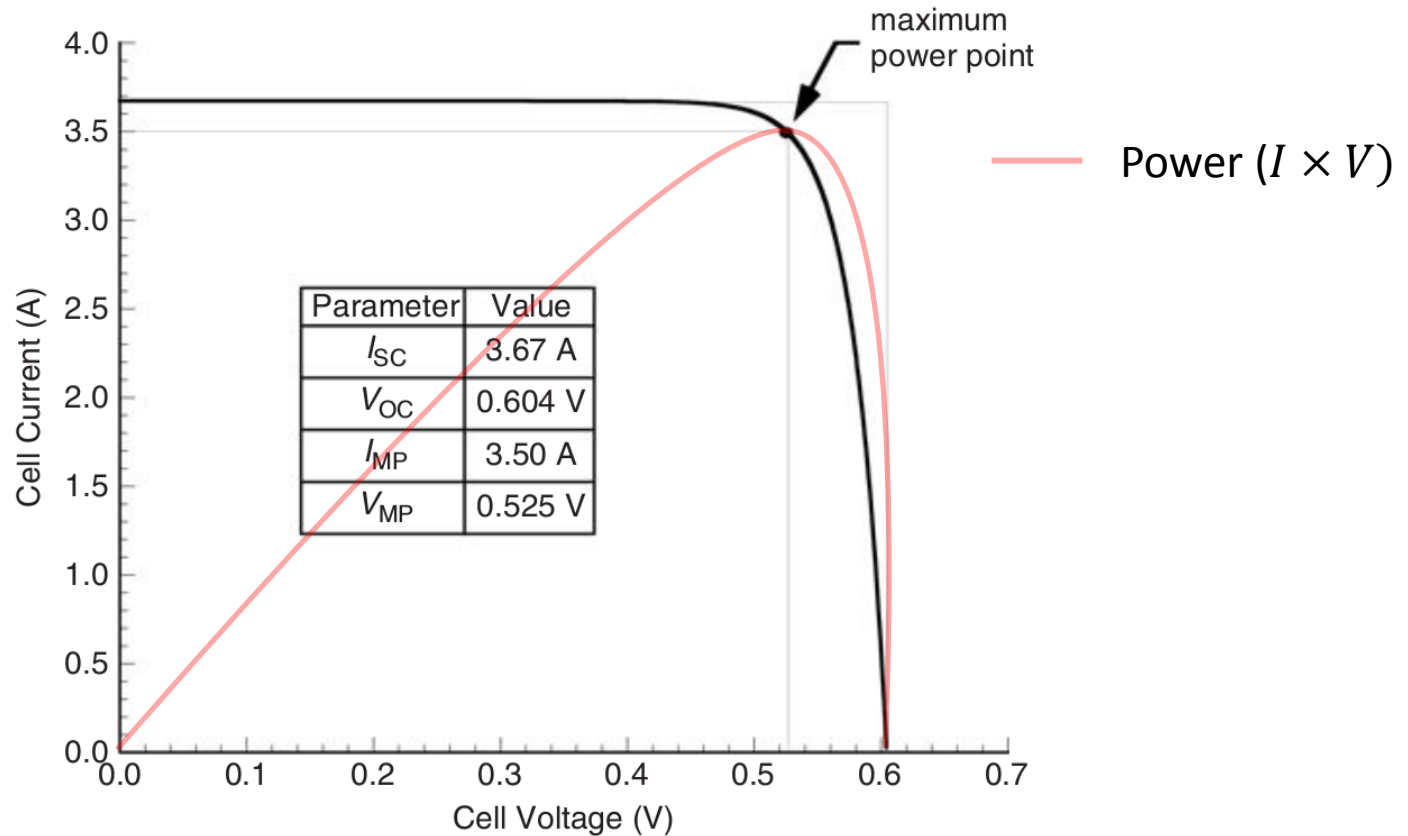


Figure 3.1 A schematic of a simple conventional solar cell. Creation of electron–hole pairs, e^- and h^+ , respectively, is depicted

Current-voltage characteristic of a solar cell

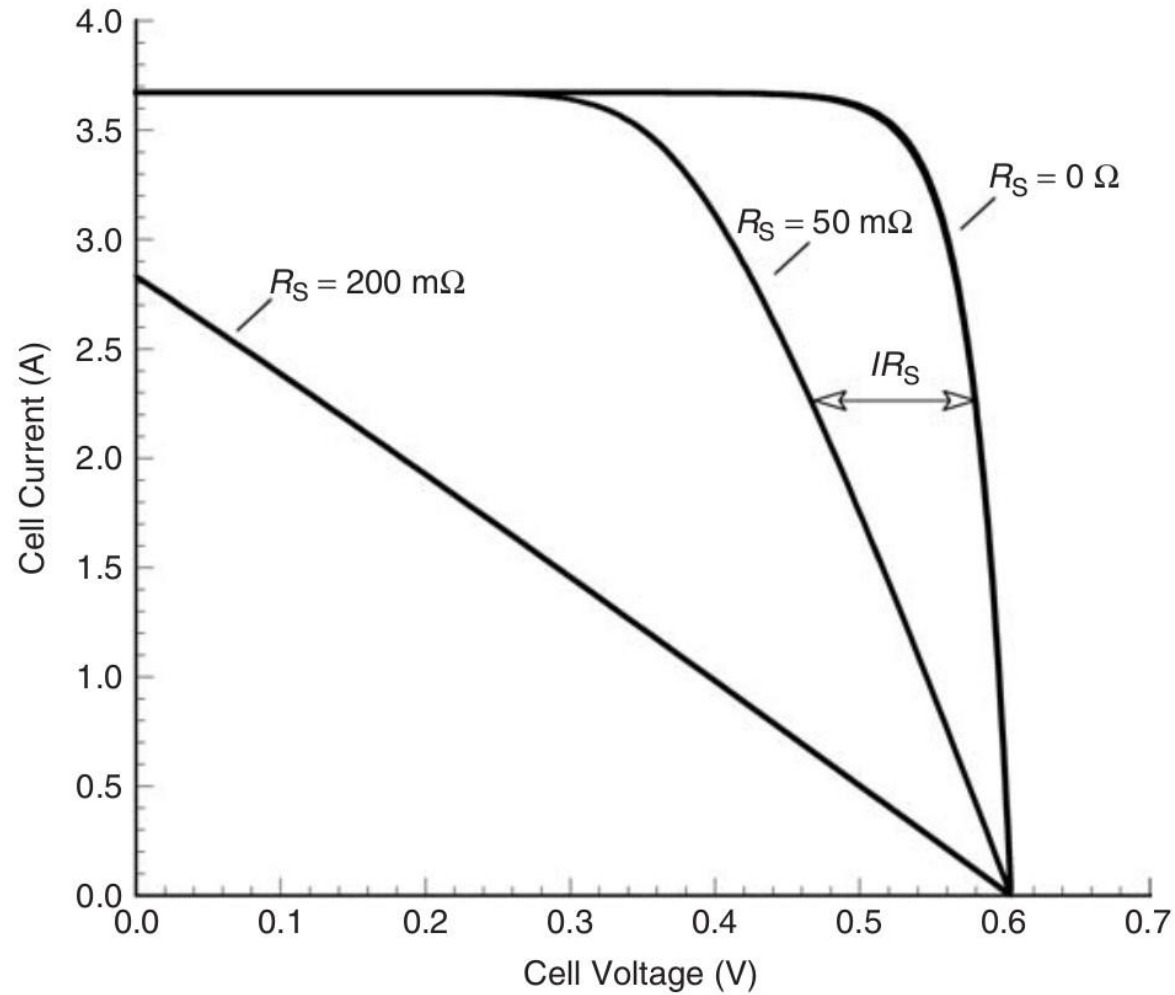


Current-voltage characteristic of a solar cell

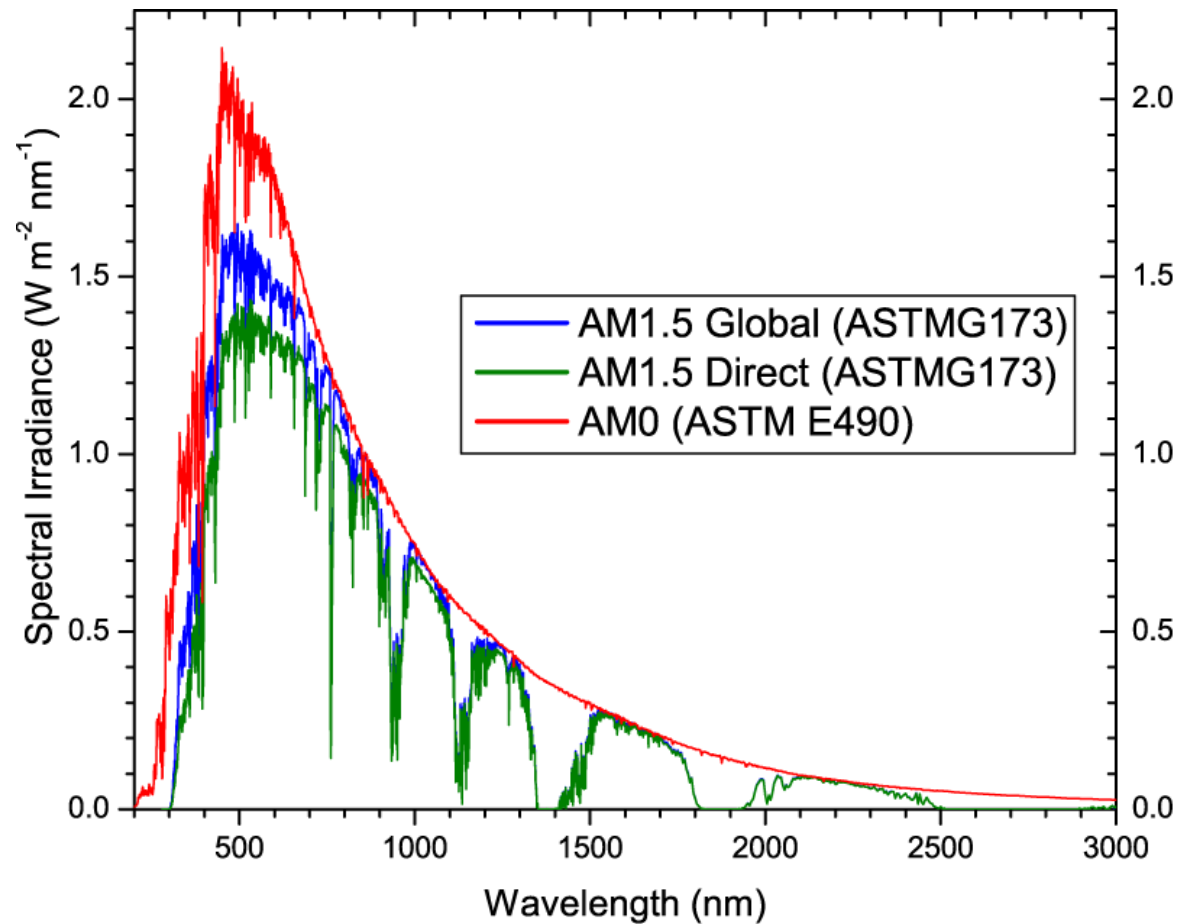


$$\eta = \frac{I_{MP} \times V_{MP}}{\text{Incident Pw}} = \frac{I_{MP} \times V_{MP} \times FF}{\text{Incident Pw}}$$

Impact of series resistance on FF

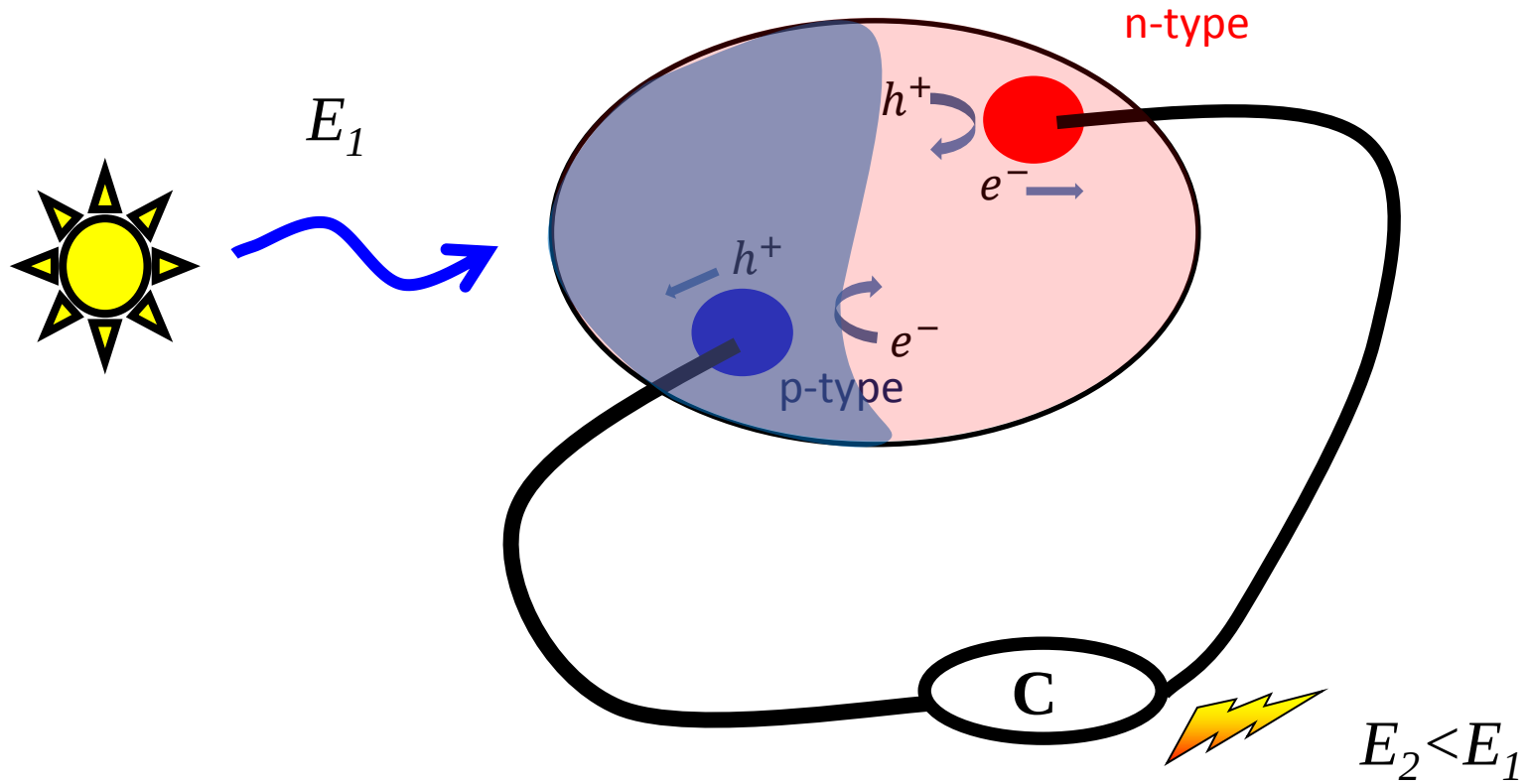


Input power (solar spectra)

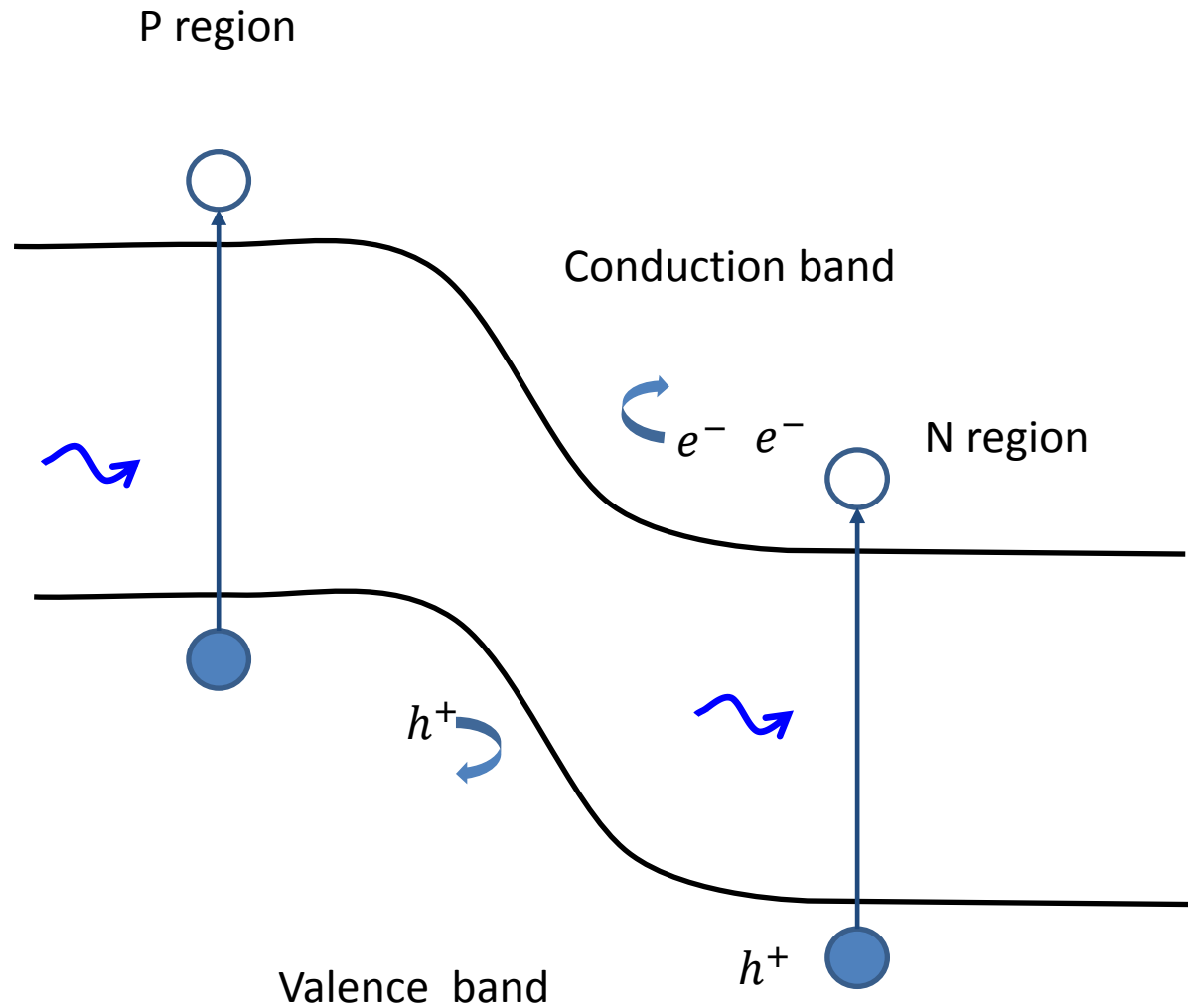


1000 Wm^{-2}
 900 Wm^{-2}
 1350 Wm^{-2}

Selective contacts: p-type and n-type semiconductors



Pn junction bandgap diagram



CURSO DE ENERGIA SOLAR FOTOVOLTAICA

Capítulo 1

Fundamentos de la Conversión Fotovoltaica: La Célula Solar

Antonio Martí
Instituto de Energía Solar - UPM

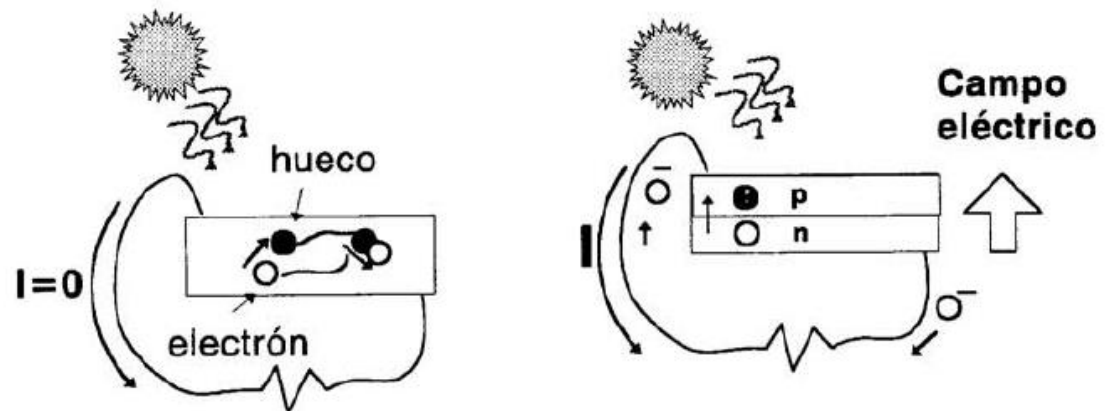
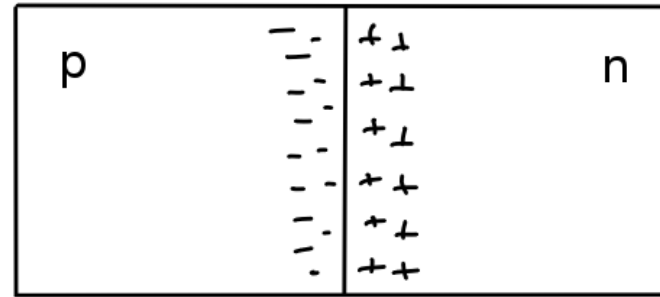


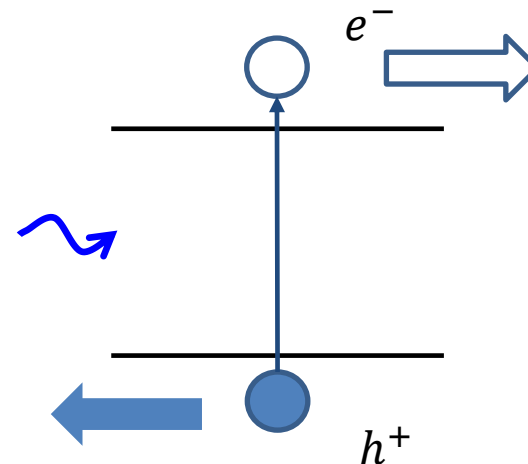
Figura 7. Un semiconductor, sin estructura *pn*, (izquierda) aunque se ilumine no provoca la circulación de corriente eléctrica. La unión *pn* hace posible la circulación de la corriente eléctrica gracias a la presencia de un campo eléctrico.

The solar cell as pn junction (wrong argument)

- p, n regions absorb light and the electric field is negligible
- Instead, the concept of electrochemical potential as driving force must be used



Electric field

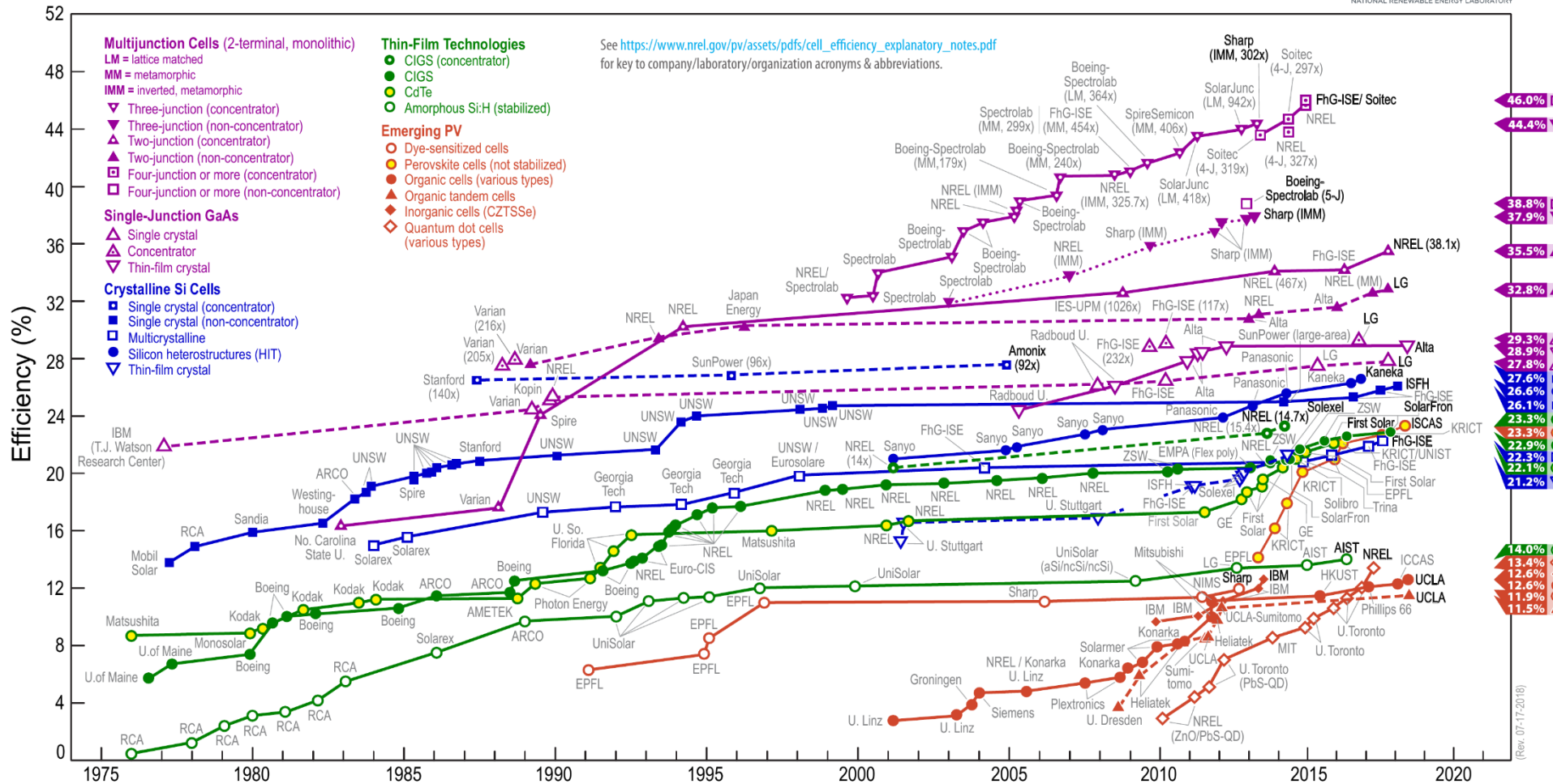


- Fundamentals of photovoltaic Energy conversion
- Conventional (inorganic) solar cells

- Fundamentals of photovoltaic Energy conversion
- Conventional (inorganic) solar cells
 - silicon
 - III-Vs (multi-junction solar cells, GaAs, InGaP, InGaAs...)
 - thin films

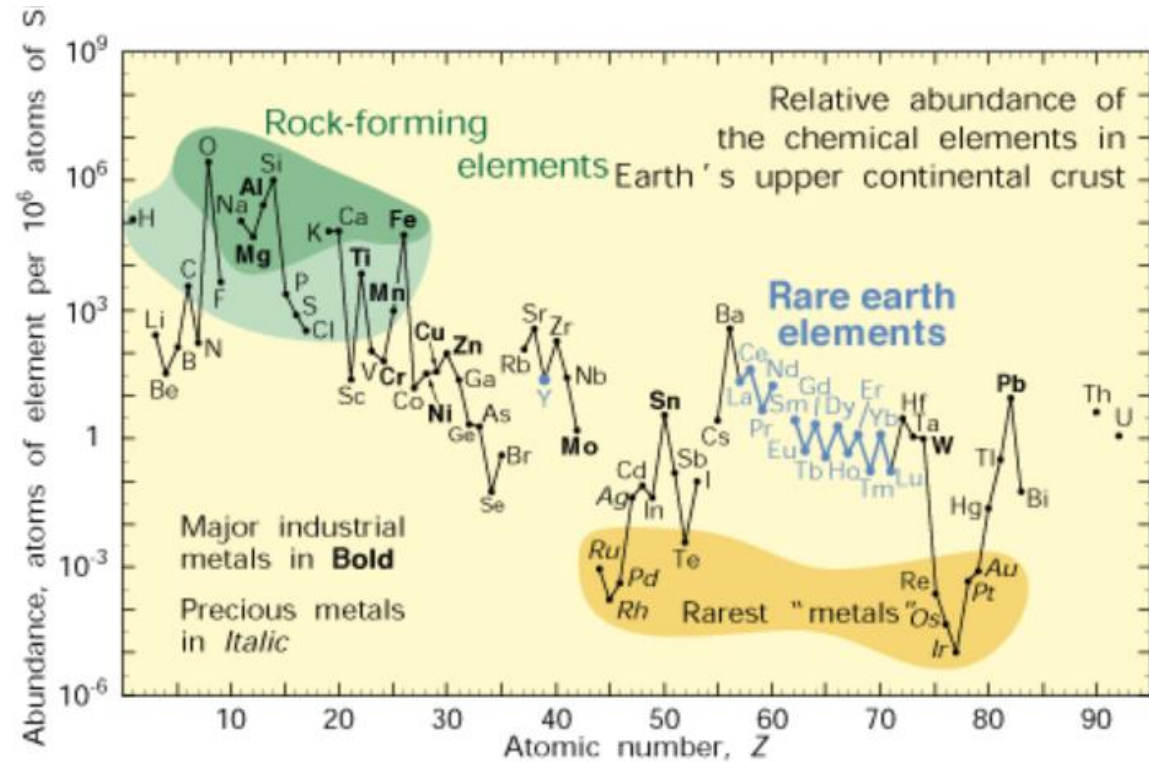
Solar Cell Efficiency records (as in 2018)

Best Research-Cell Efficiencies



The abundance of materials problem

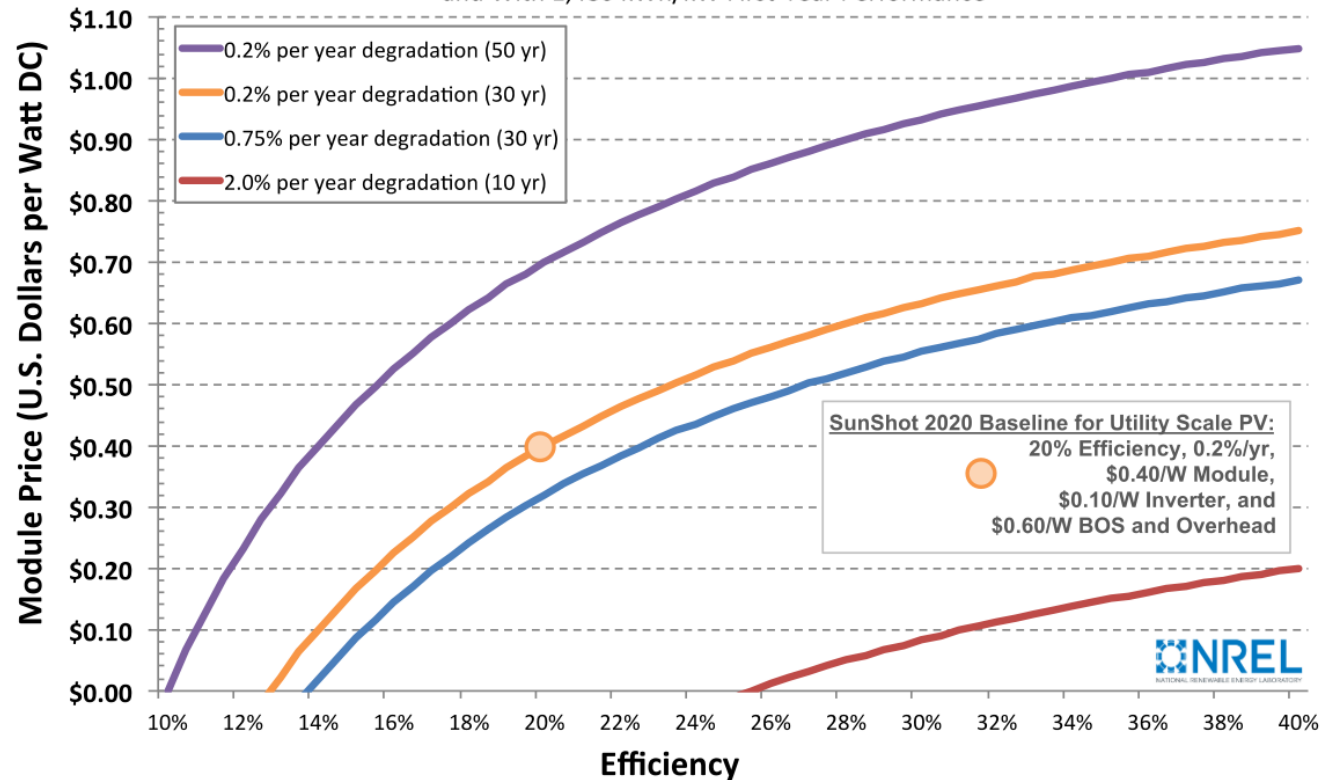
- Si – not a problem
- $\text{Cu(InGa)Se}_2 \rightarrow \text{In}$
- $\text{CdTe} \rightarrow \text{Te}$
- Multijunction $\rightarrow \text{Ge}$



P.H. Stauffer et al, Rare Earth Elements - Critical Resources for High Technology, USGS (2002)

Metric Sets to Achieve the Utility Scale SunShot Goal

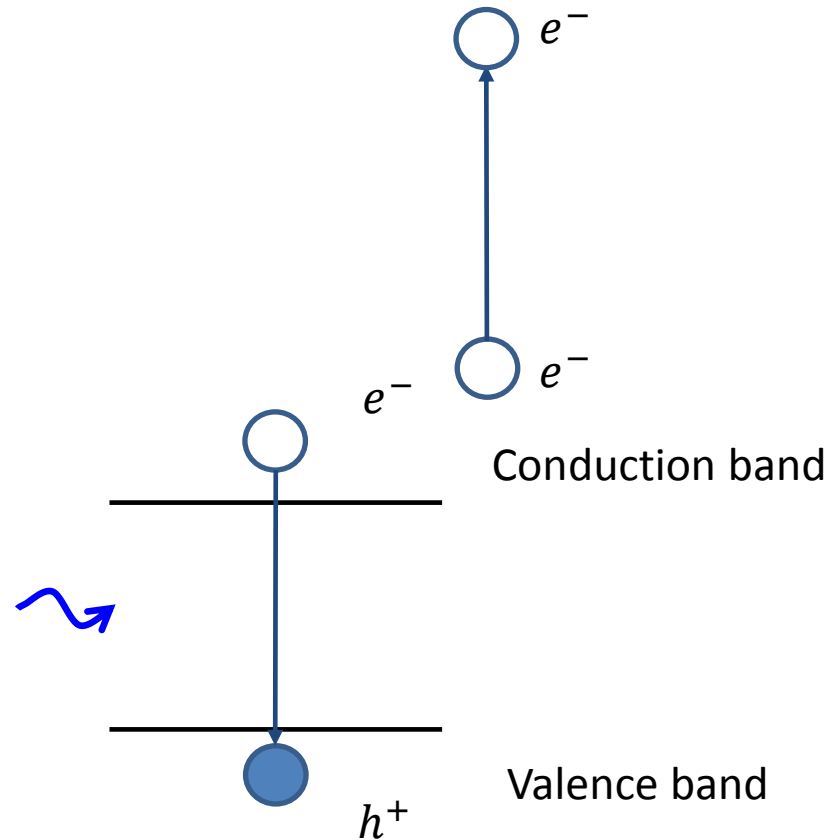
Iso-LCOE Curves of 6 cents per kWh Without Federal or State Incentives
and With 1,480 kWh/kW First-Year Performance



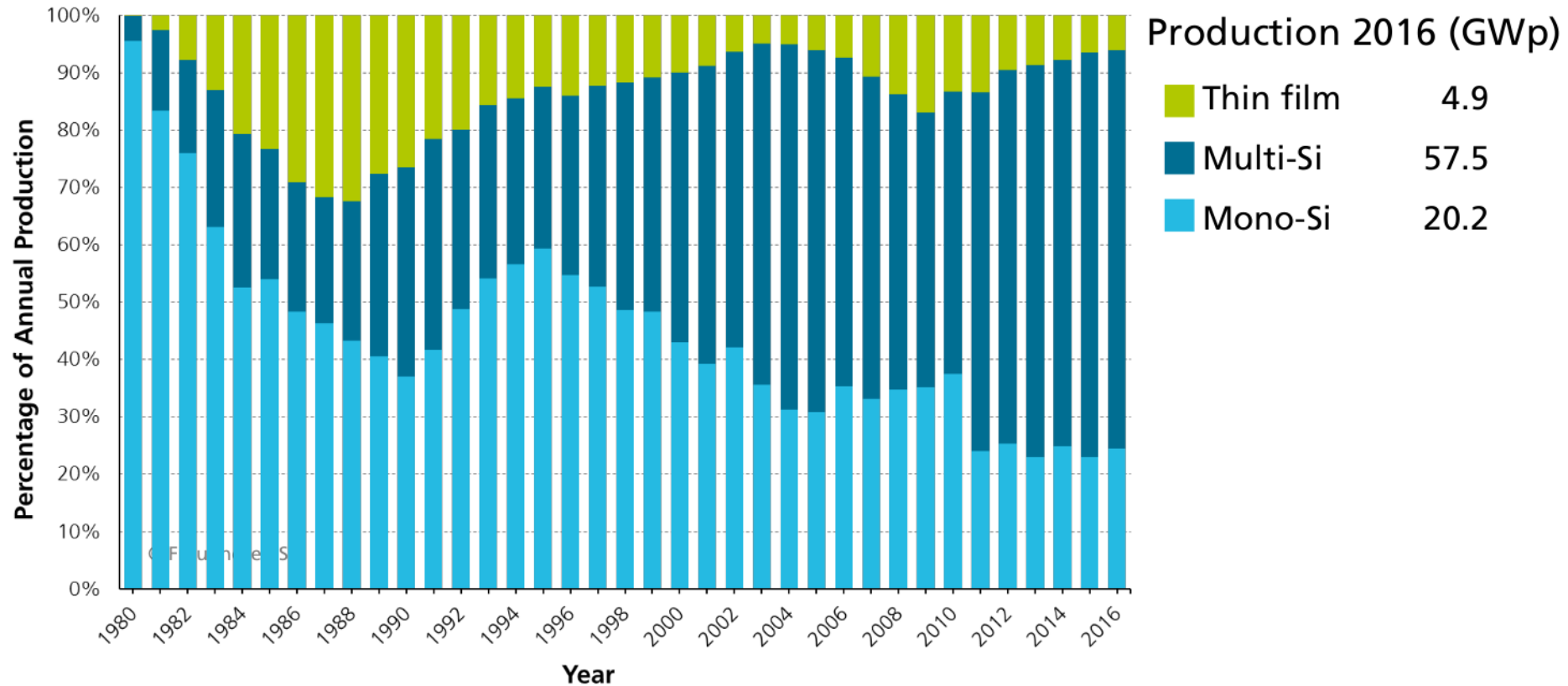
- Fundamentals of photovoltaic Energy conversion
- Conventional (inorganic) solar cells
 - silicon
 - III-Vs (multi-junction solar cells, GaAs, InGaP, InGaAs...)
 - thin films

Silicon: some properties

- Weak absorption
- Recombination limited by Auger



Silicon dominates de market



Monocrystalline and Multicrystalline modules

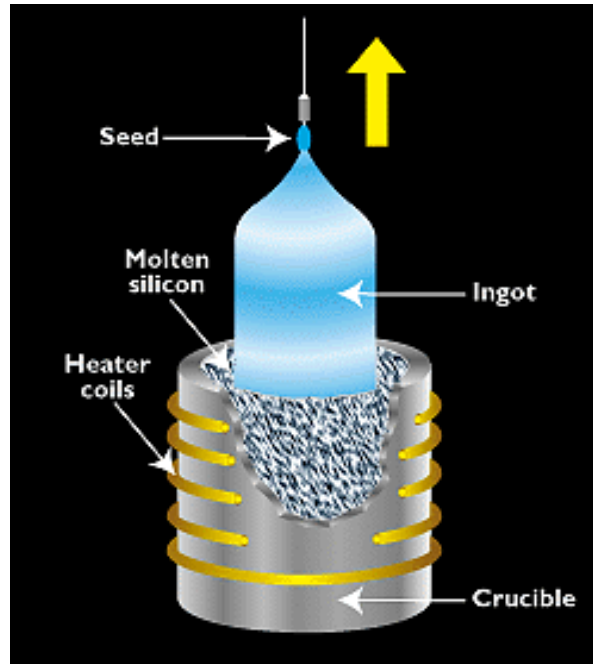


Mono

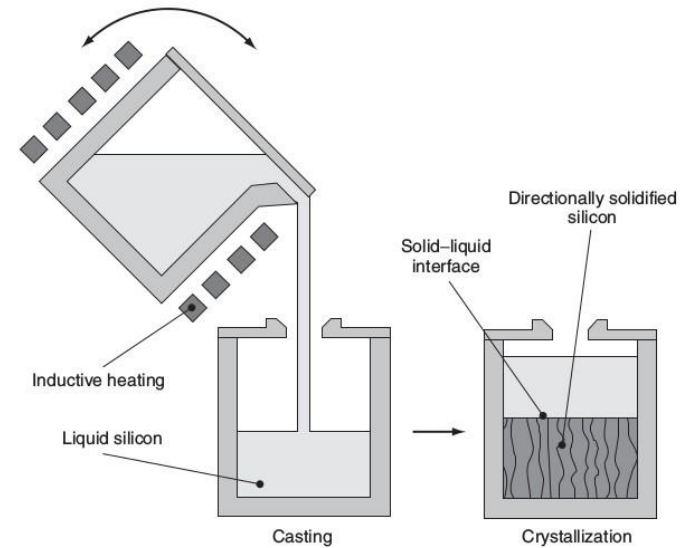


Multi

Monocrystalline and Multicrystalline modules



Mono



Multi



Russel Ohl
(Technology Review)

June 25, 1946.

R. S. OHL

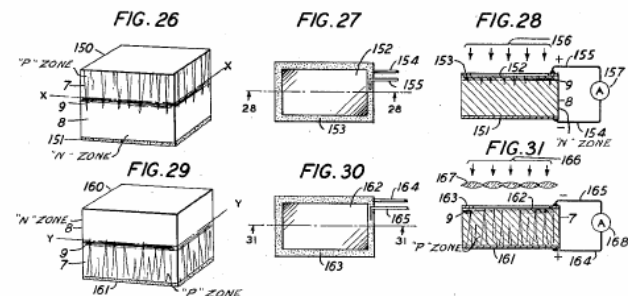
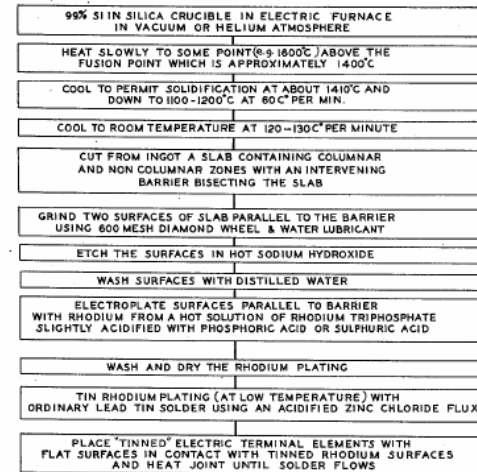
2,402,662

LIGHT-SENSITIVE ELECTRIC DEVICE

Filed May 27, 1941

5 Sheets-Sheet 3

FIG. 14

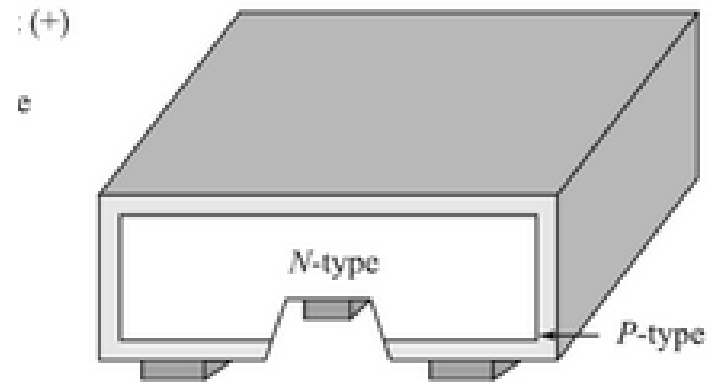

 INVENTOR
 R. S. OHL

 BY
 Stanley B. Kent
 ATTORNEY

1954 (6%)



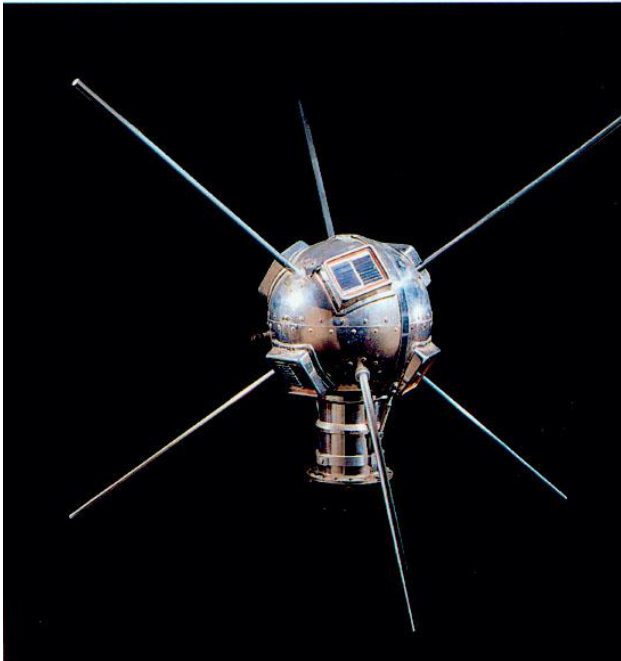
Person, Chapin,
Fuller (Perlin, The
silicon solar cell
turns 50)



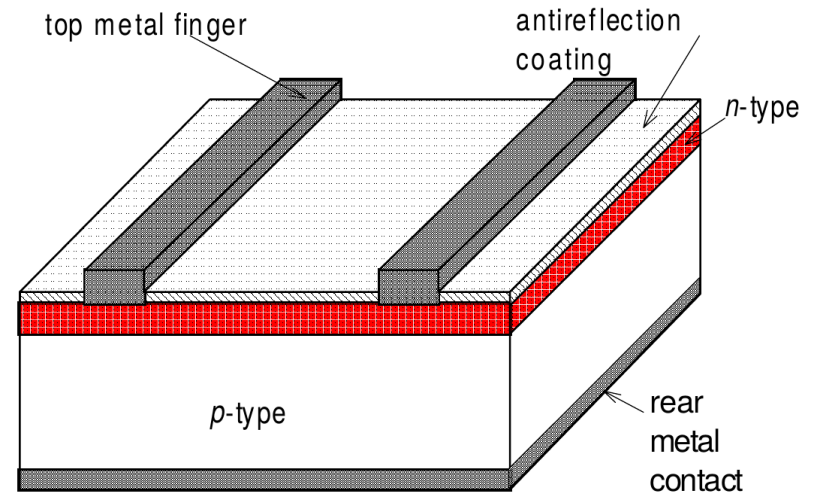
b) Structure of 6% efficient solar cell in CZSi wafer
and diffused junction developed in 1954



1956 advertisement of “Look magazine” (Perlin, The silicon solar cell turns 50)

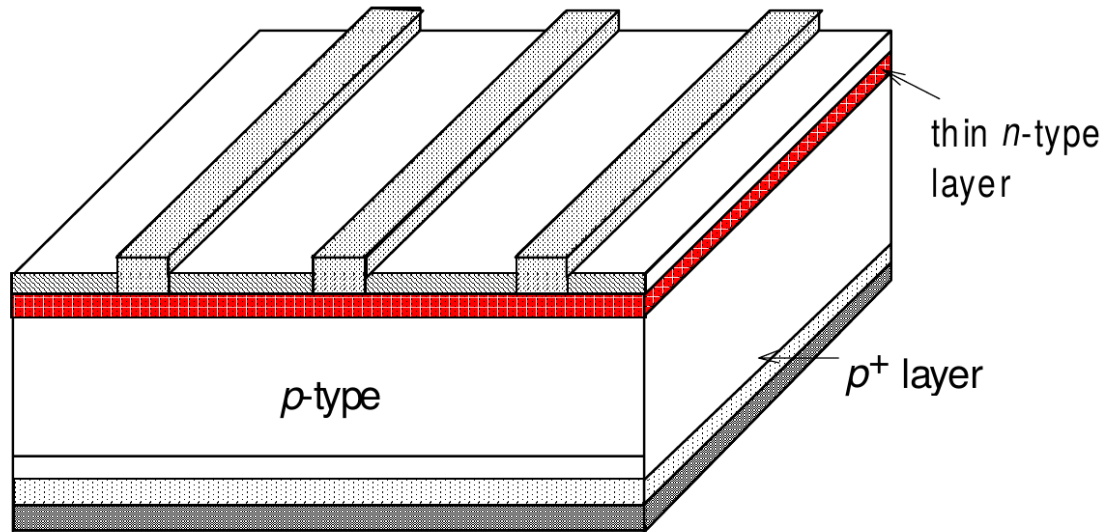


Vanguard I (1958)



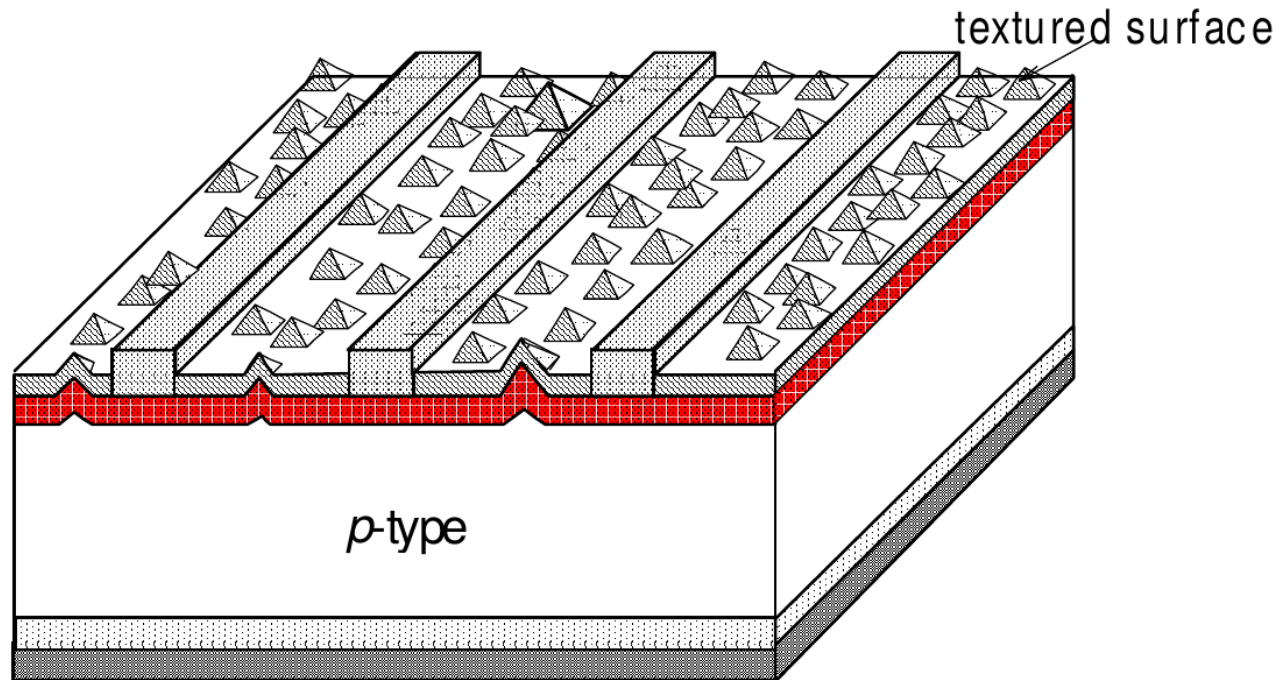
(Source: M.A.Green, Chap 4 in Clean Energy from Photovoltaics)

1972 – 15 % (Violet cell)

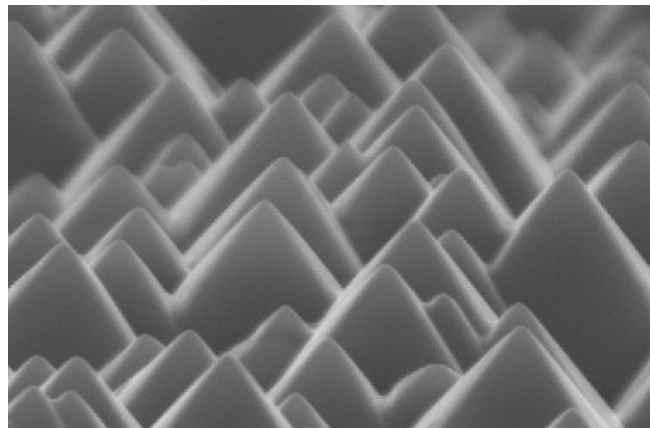
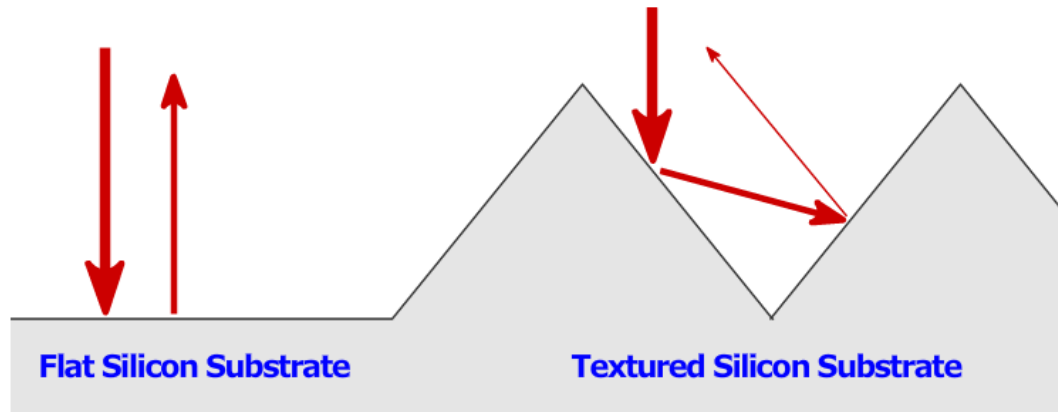


(Source: M.A.Green, Chap 4 in Clean Energy from Photovoltaics)

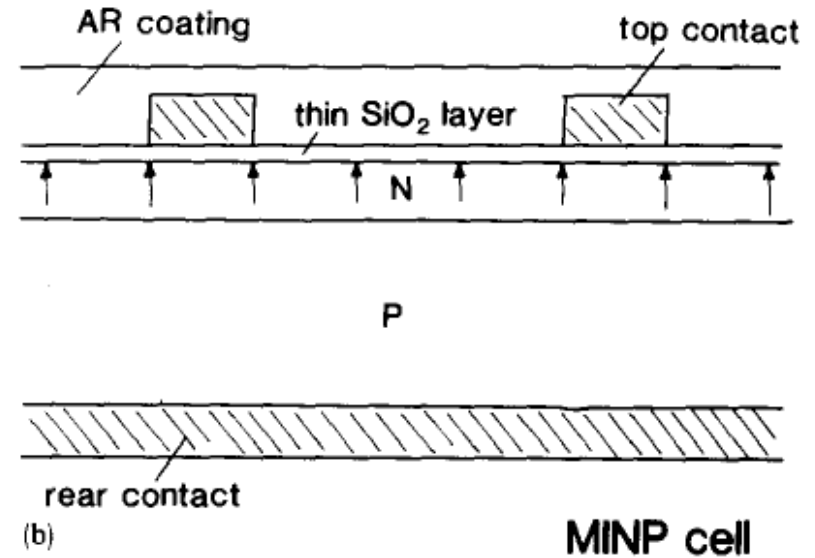
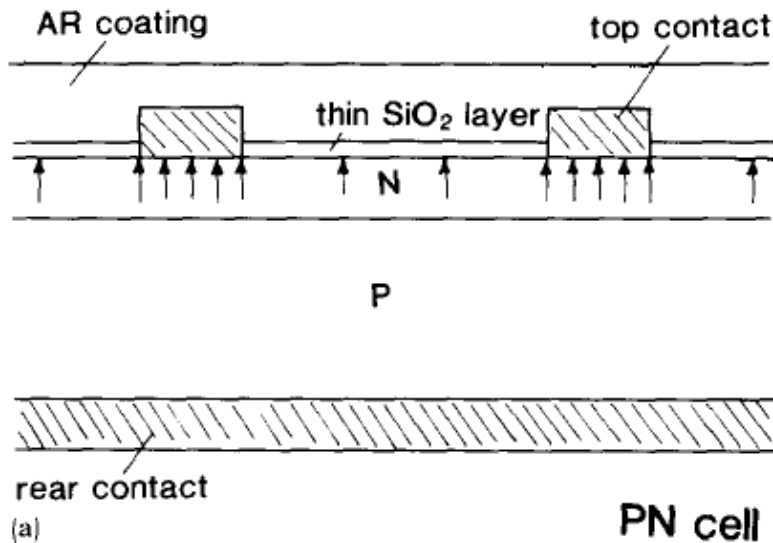
1974 – 18% (Black cell)



(Source: M.A.Green, Chap 4 in Clean Energy from Photovoltaics)

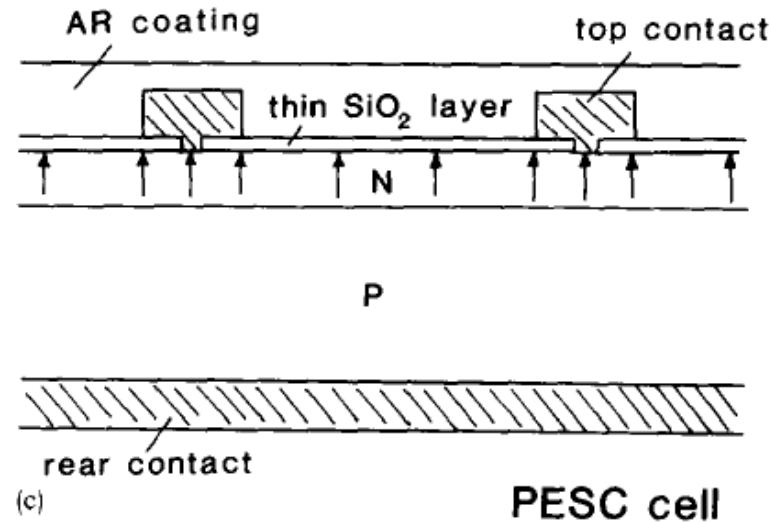


1983 – 18% (Metal to insulator np junction - MINP cell)



Green, M. A., Blakers, A. W., Shi, J., Keller, E. M., & Wenham, S. R. (1984). 19.1% efficient silicon solar cell. *Applied Physics Letters*, 44(12), 1163-1164.

1984 – 19% (Passivated emitter solar cell – PESCS)



Green, M. A., Blakers, A. W., Shi, J., Keller, E. M., & Wenham, S. R. (1984). 19.1% efficient silicon solar cell. *Applied Physics Letters*, 44(12), 1163-1164.

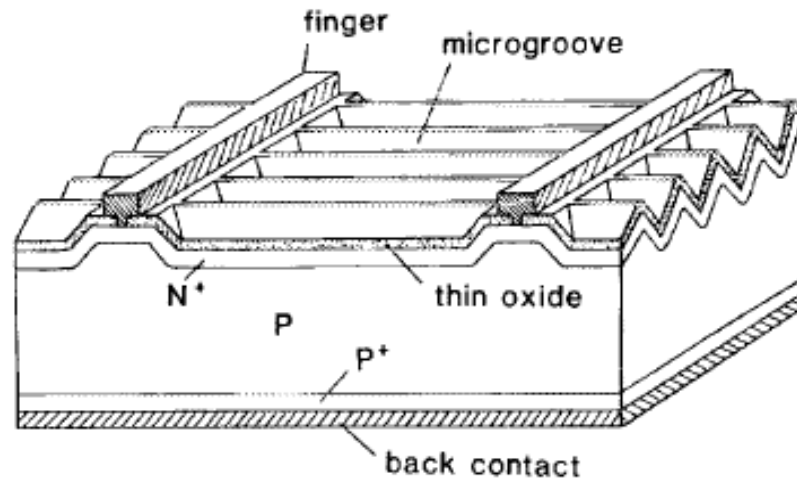
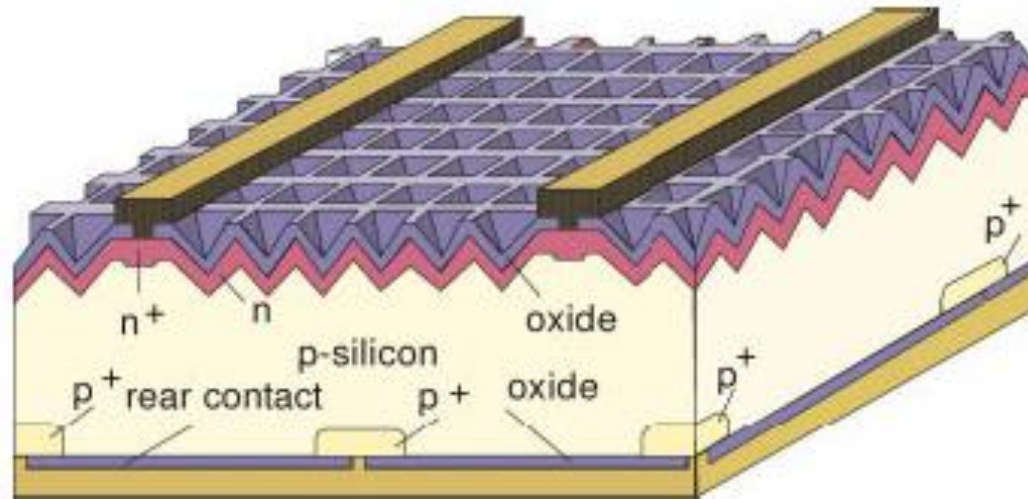


FIG. 1. Schematic diagram of a microgrooved passivated emitter solar cell (PESC). The cell double layer antireflection coating is not shown (not to scale).

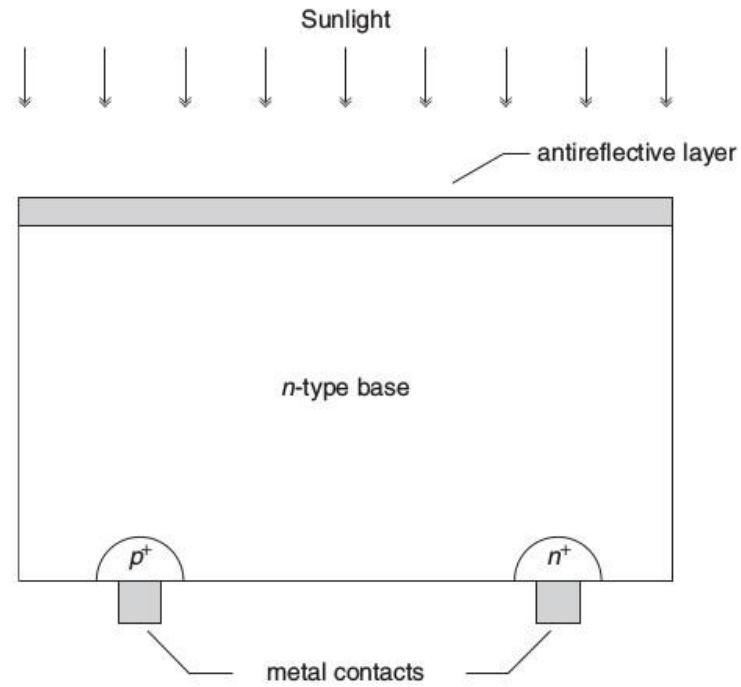
Blakers, A. W., & Green, M. A. (1986). 20% efficiency silicon solar cells. *Applied physics letters*, 48(3), 215-217.

2009 – 25 % (Passivated emitter locally diffused PERL cell)

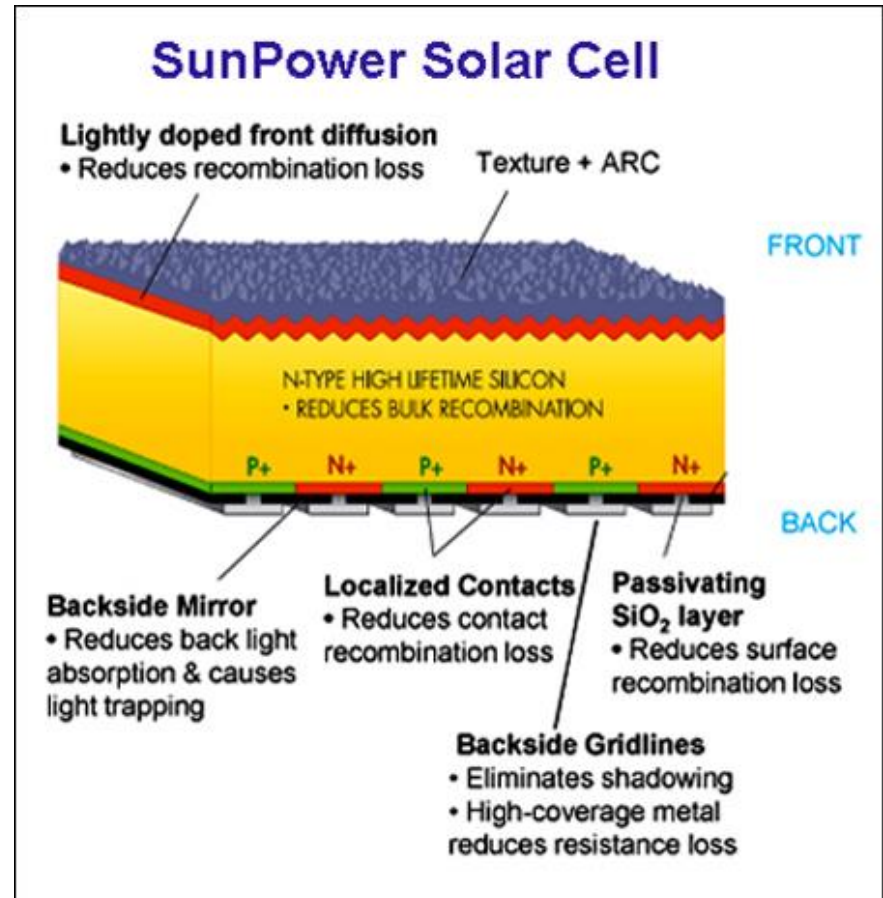


Green, M. A. (2009). The path to 25% silicon solar cell efficiency: History of silicon cell evolution. Progress in Photovoltaics: Research and Applications, 17(3), 183-189.

1990 – 22 % (back contact, rear junction solar cell)

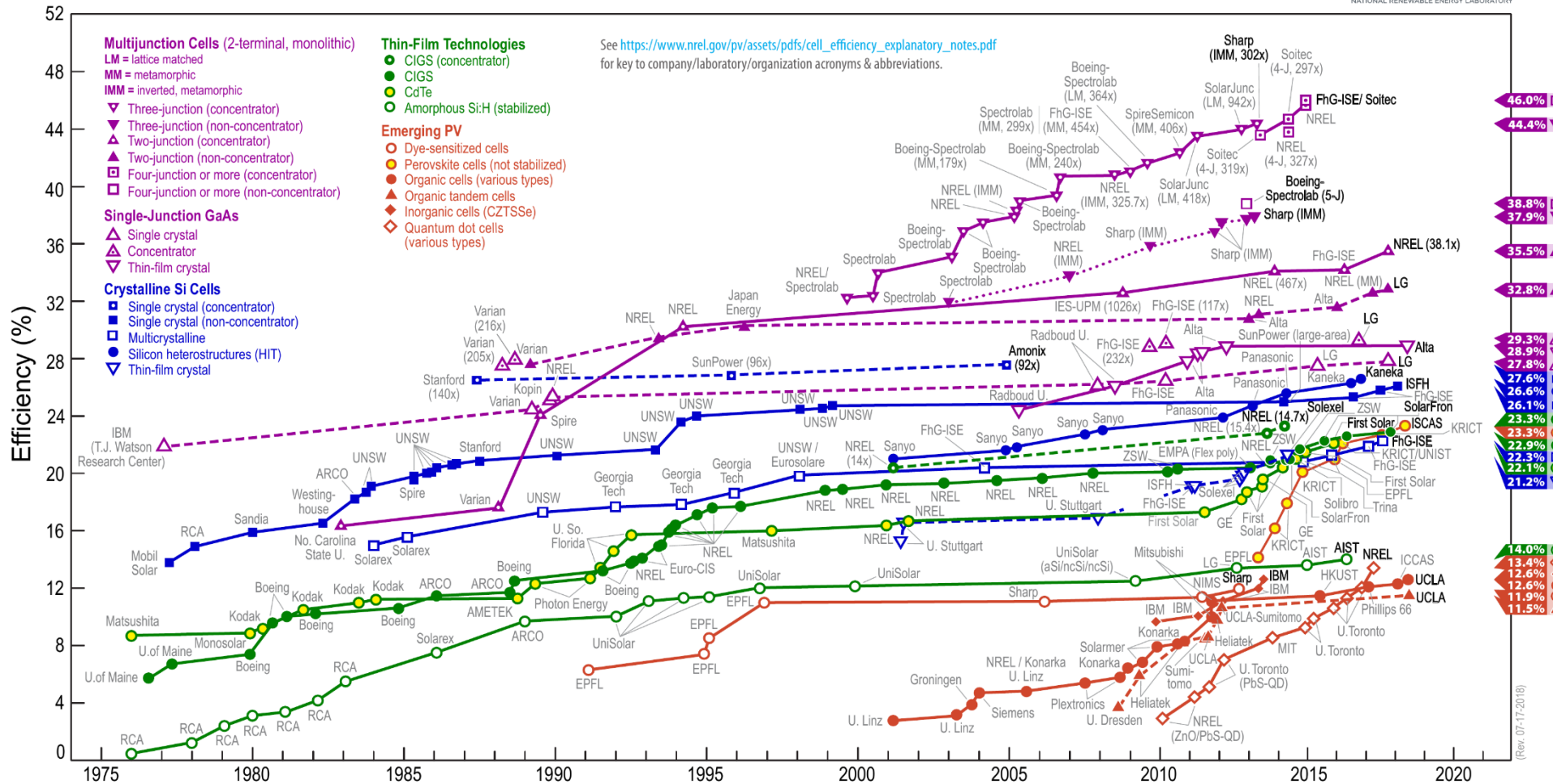


Back contact, rear junction cell (commercial)

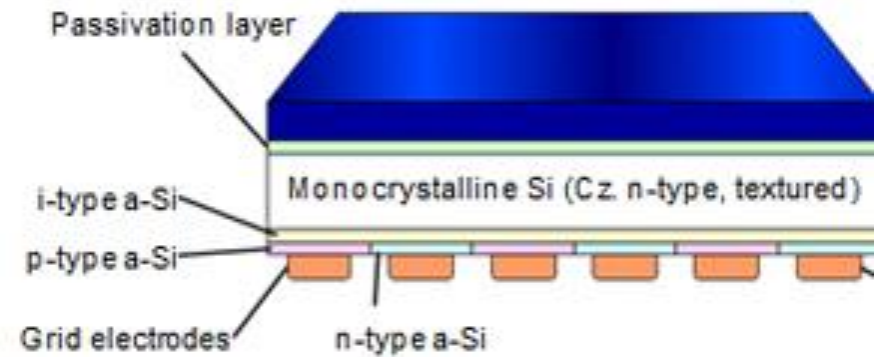


Solar Cell Efficiency records (as in 2018)

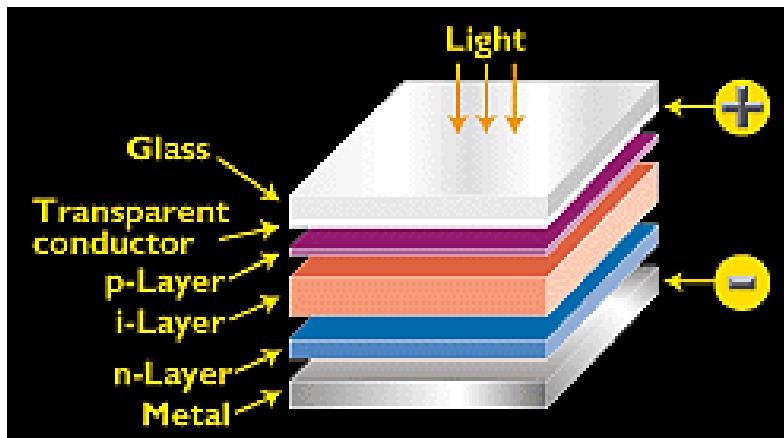
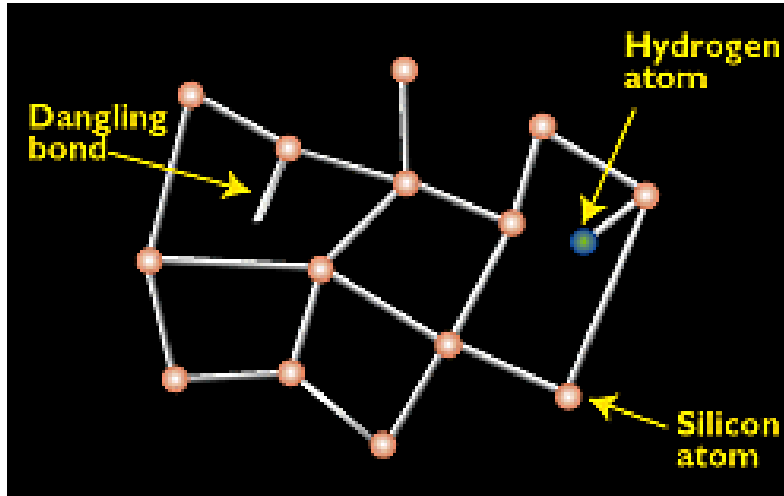
Best Research-Cell Efficiencies



TCO $\Phi = 5.3 \text{ eV}$
aSi:H(p+) Thickness= 10 nm
aSi:H(i) Thickness= 10 nm
cSi(n) 1-5 $\Omega\text{-cm}$ Thickness= 100 μm
aSi:H(i) Thickness= 10 nm
aSi:H(n+) Thickness= 10 nm
Al

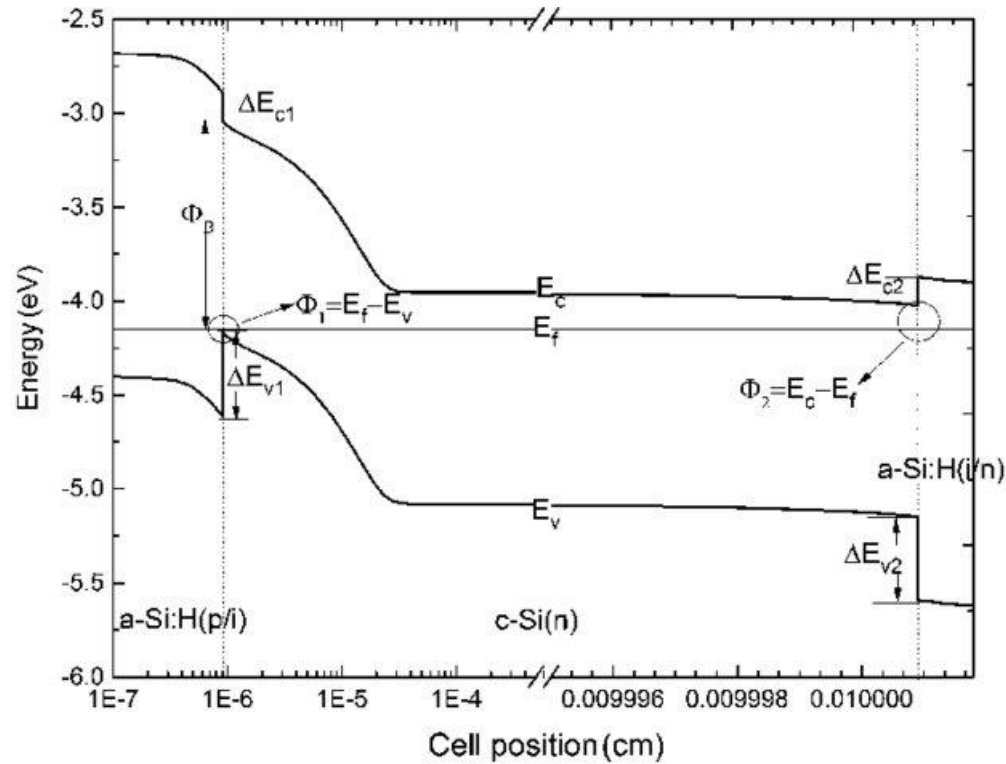


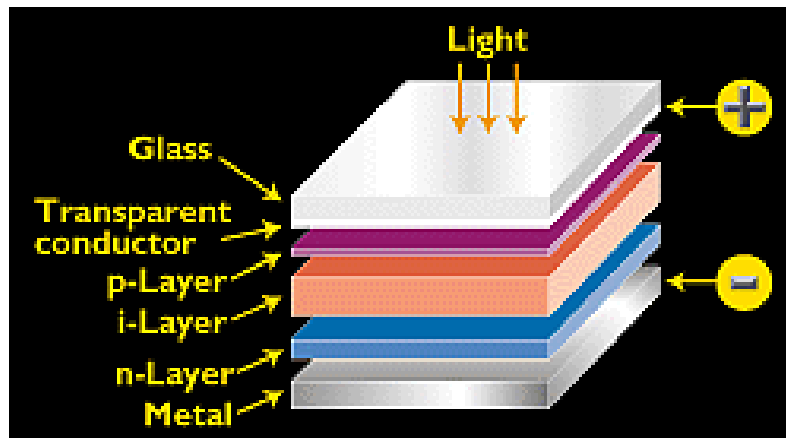
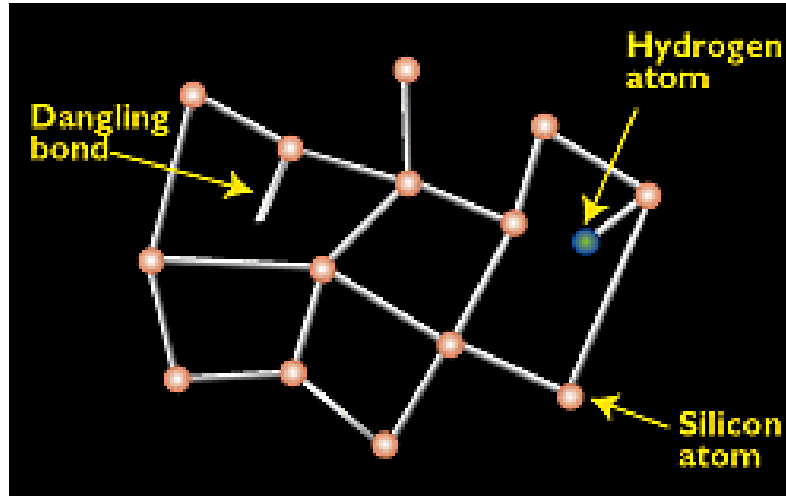
Panasonic



- Hydrogenated amorphous silicon: a-Si:H
- Greater absorption (thinner cells)
- Fabricated by CVD technology (RF PECVD)
- Degradation problems
- Tunable bandgap (1,7 eV):
 - with Ge, decreases (1.45 eV)
 - with C,N increases (2 eV)
- Possibility of tandem solar cells

Band diagram of a HIT cell





- Hydrogenated amorphous silicon: a-Si:H
- Greater absorption (thinner cells)
- Fabricated by CVD technology (RF PECVD)
- Degradation problems
- Tunable bandgap (1,7 eV):
 - with Ge, decreases (1.45 eV)
 - with C,N increases (2 eV)
- Possibility of tandem solar cells

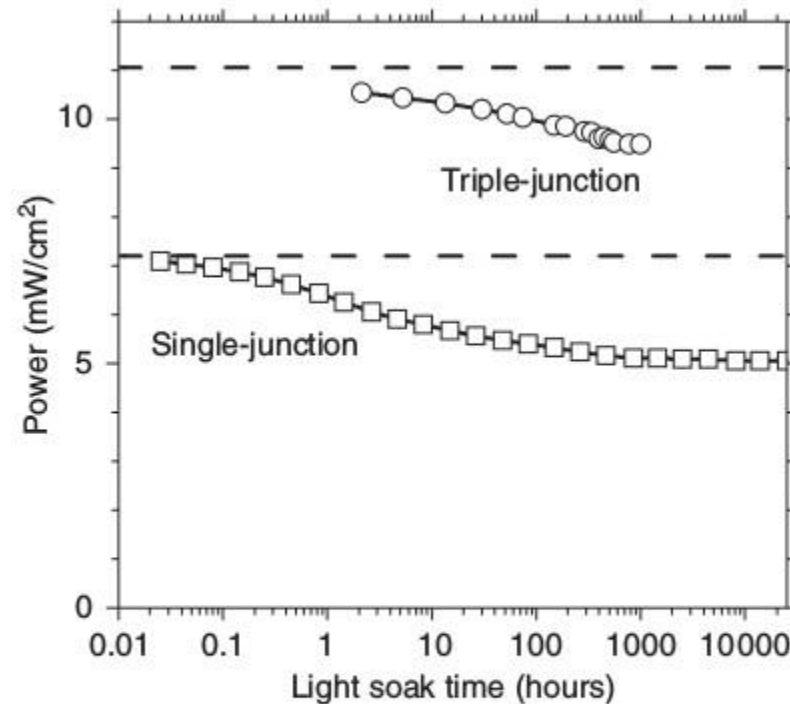
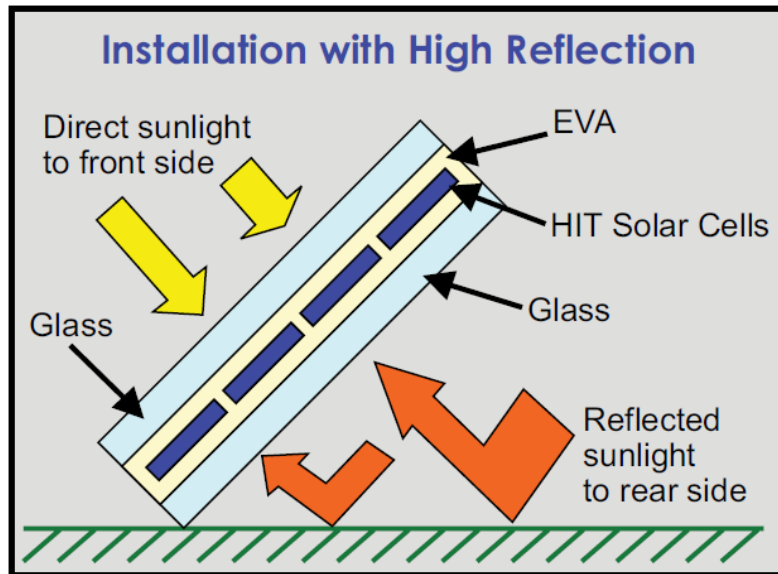


Figure 12.4 The conversion efficiency in a-Si:H-based solar cells declines noticeably upon the first exposure to sunlight. The figure illustrates this decline under a solar simulator (100 mW/cm²) for a single-junction cell (260 nm *i*-layer thickness) and for a triple junction module made at United Solar Ovonic [16, 17]; the dashed lines indicate the initial power measured for each device



Source: Sanyo Energy Corporation

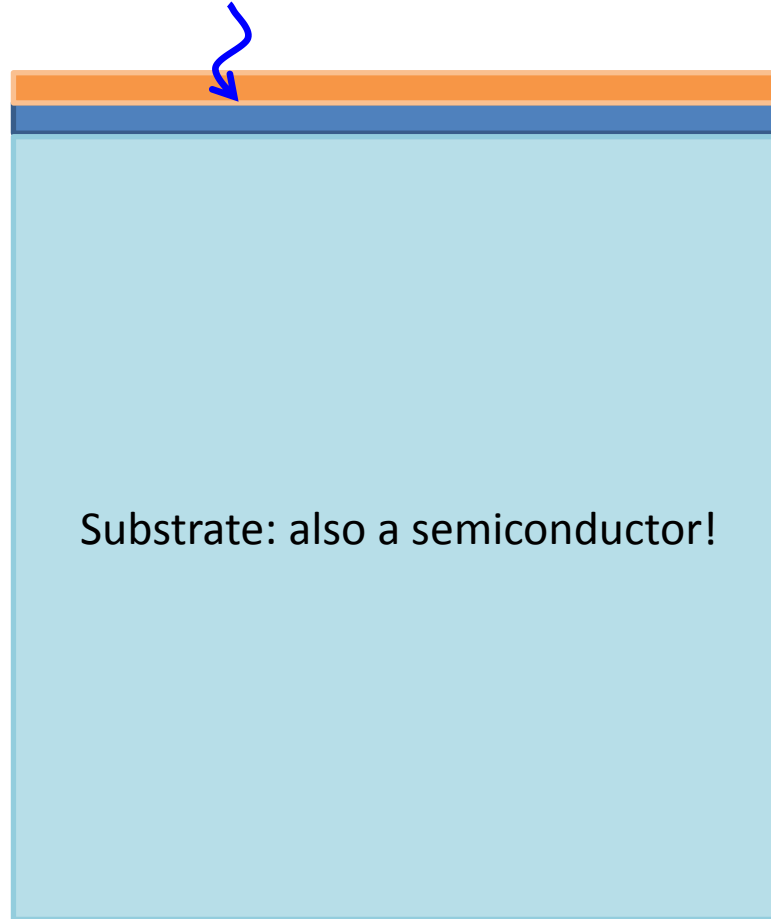


Source: Silfab (Oregon park)

- Fundamentals of photovoltaic Energy conversion
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 - thin films

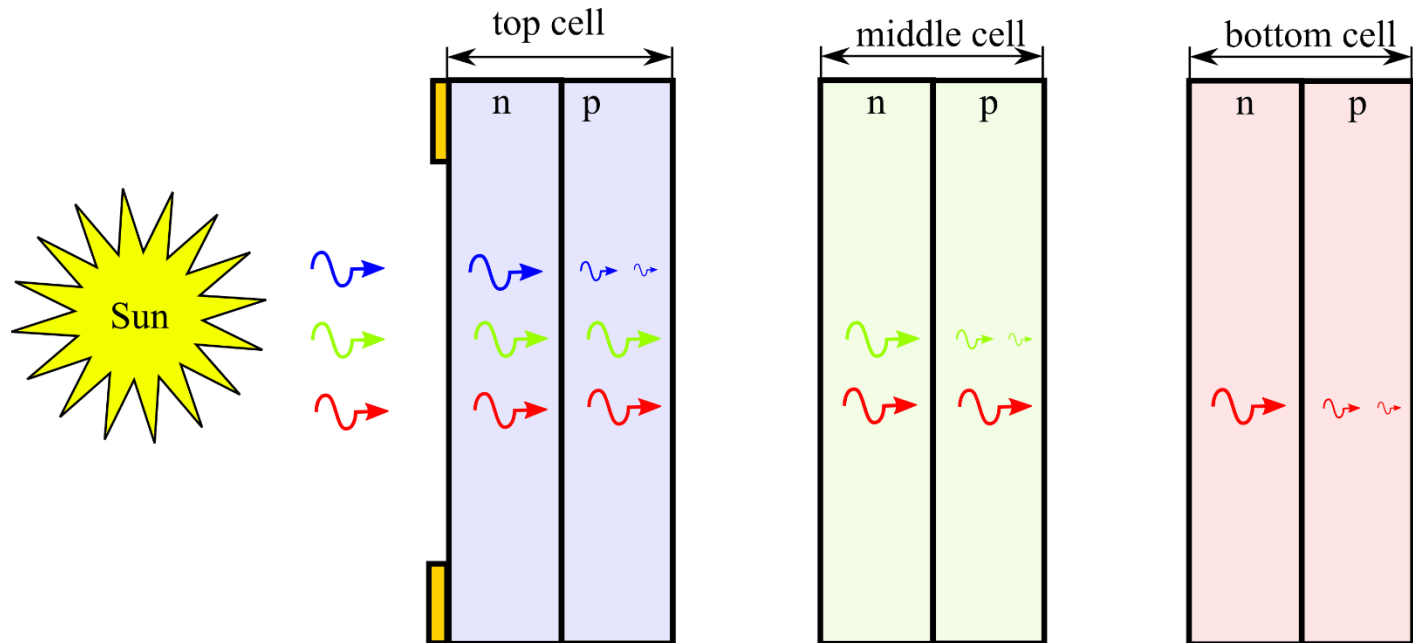
III-Vs: some properties

- Strong absorption
- Difficult to stack
- Work in the radiative limit
- Today driving market are space applications

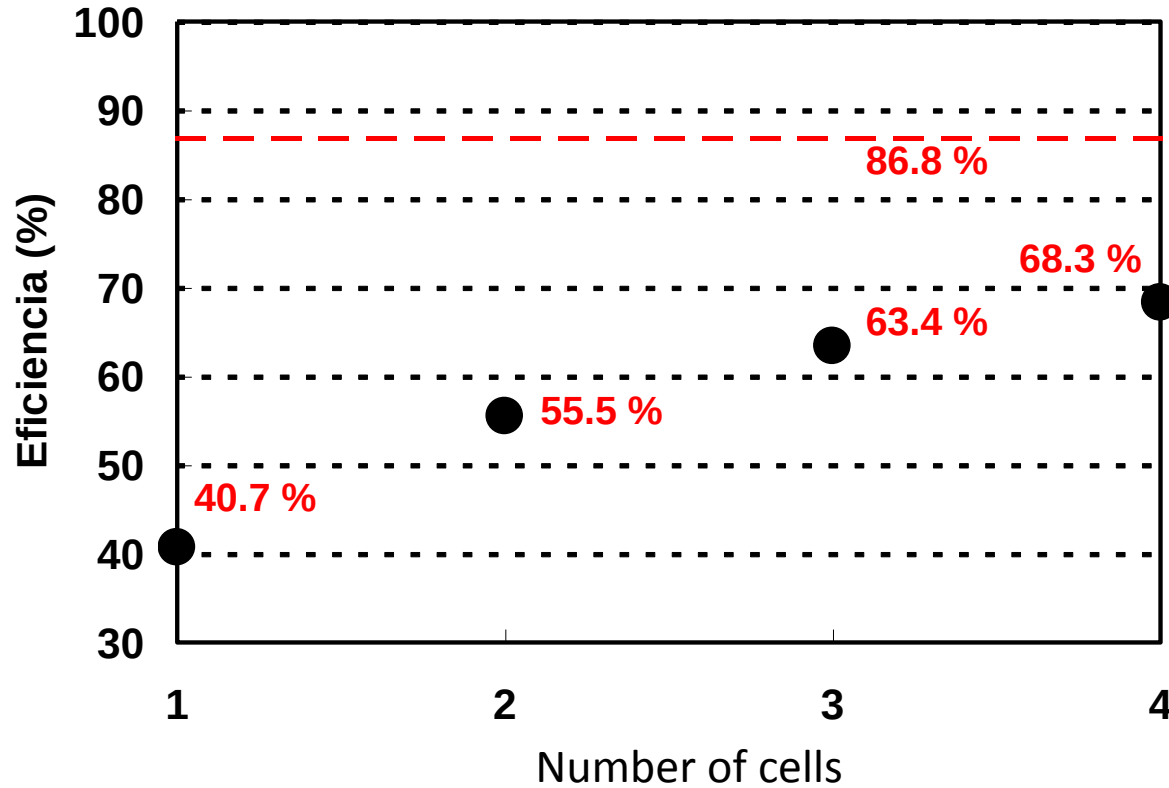


P-n junction

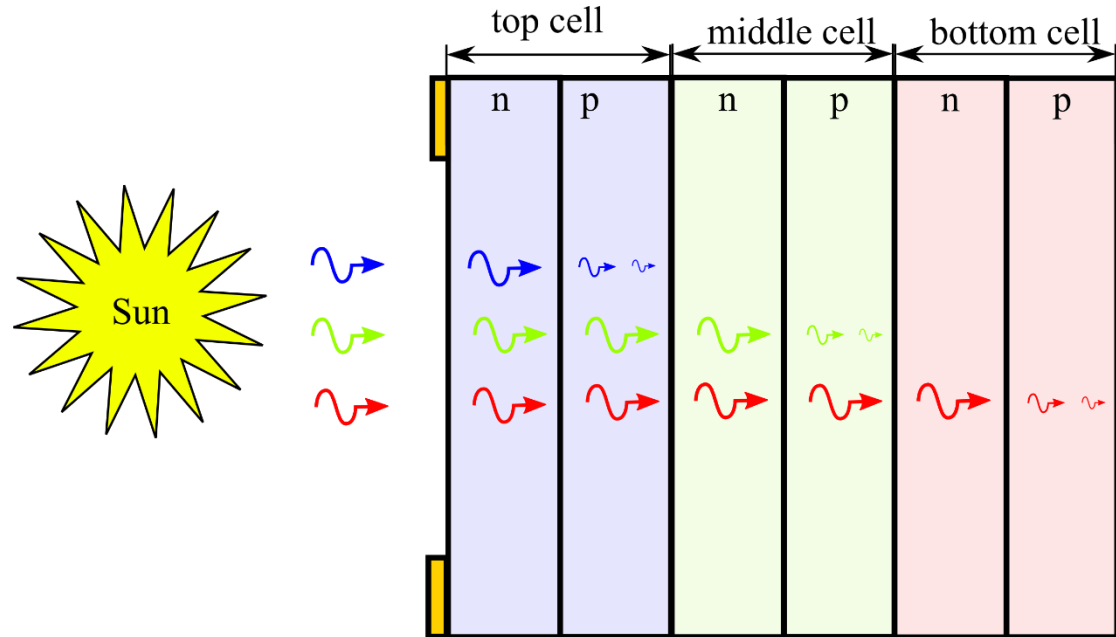
Tandem cells



Tandem cells: Limiting efficiency

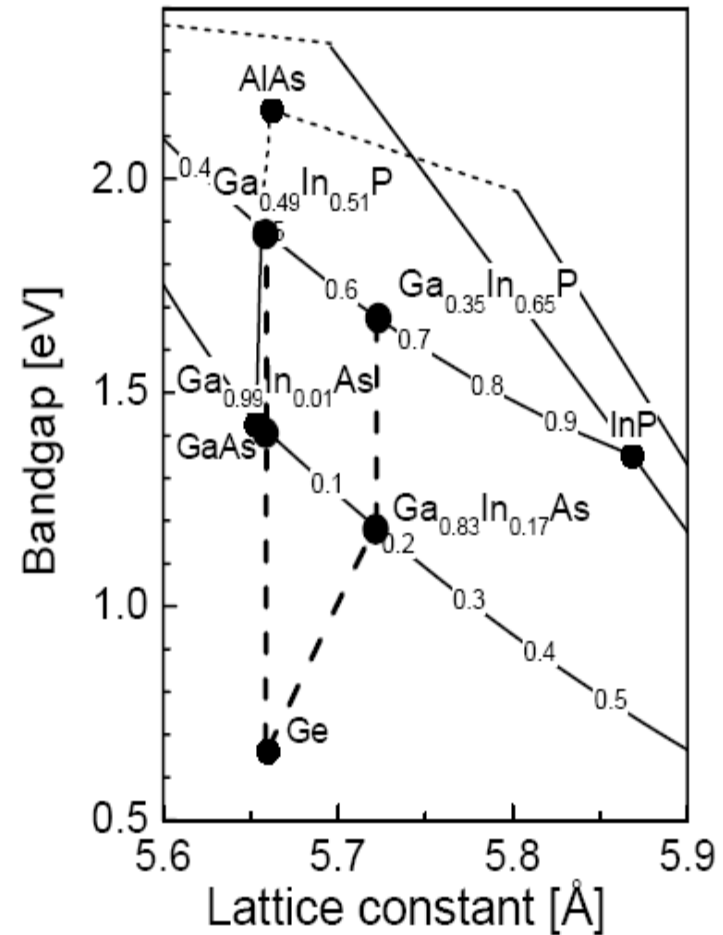
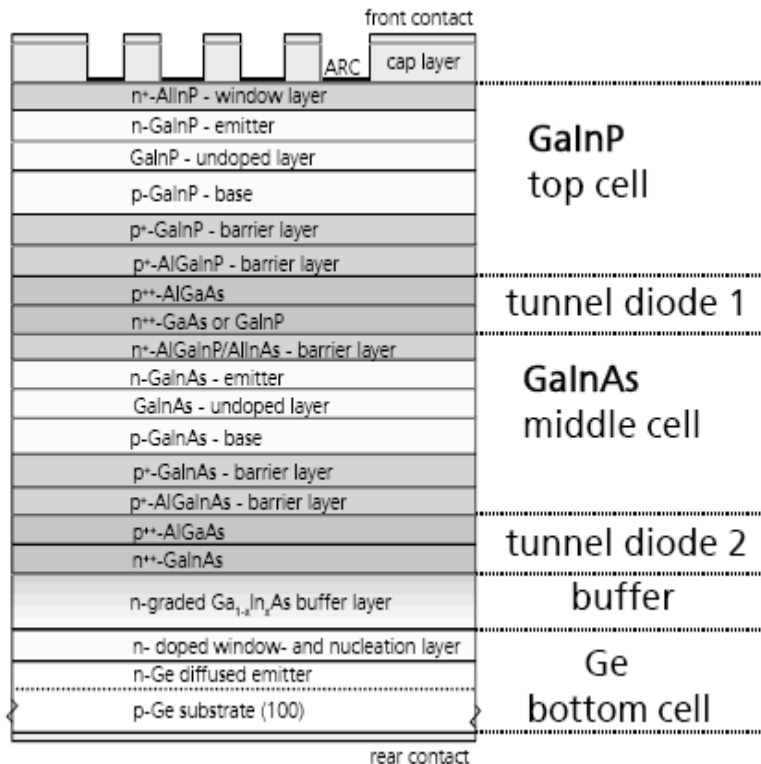


Tandem cells: conexión en serie



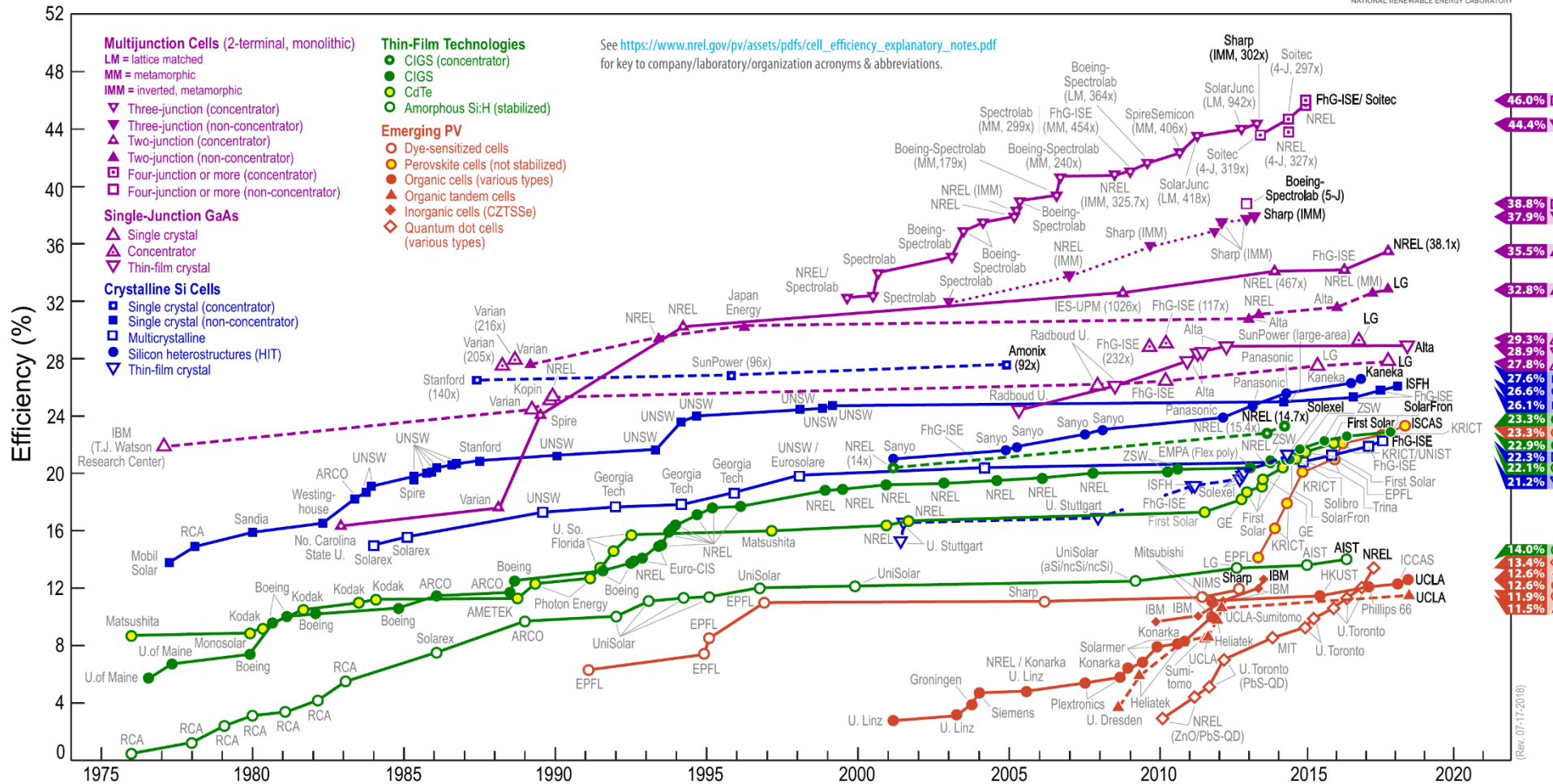
The current has to be the same for all the cells

Multi-junction solar cells: lattice matched and methamorphic



Solar Cell Efficiency records (as in 2018)

Best Research-Cell Efficiencies



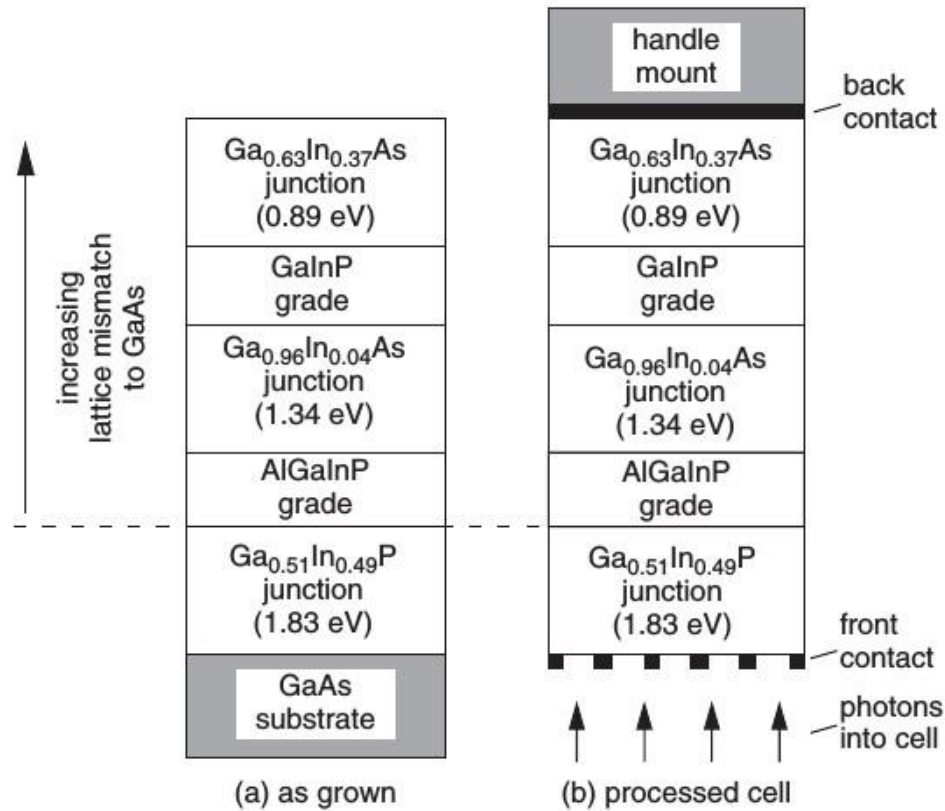
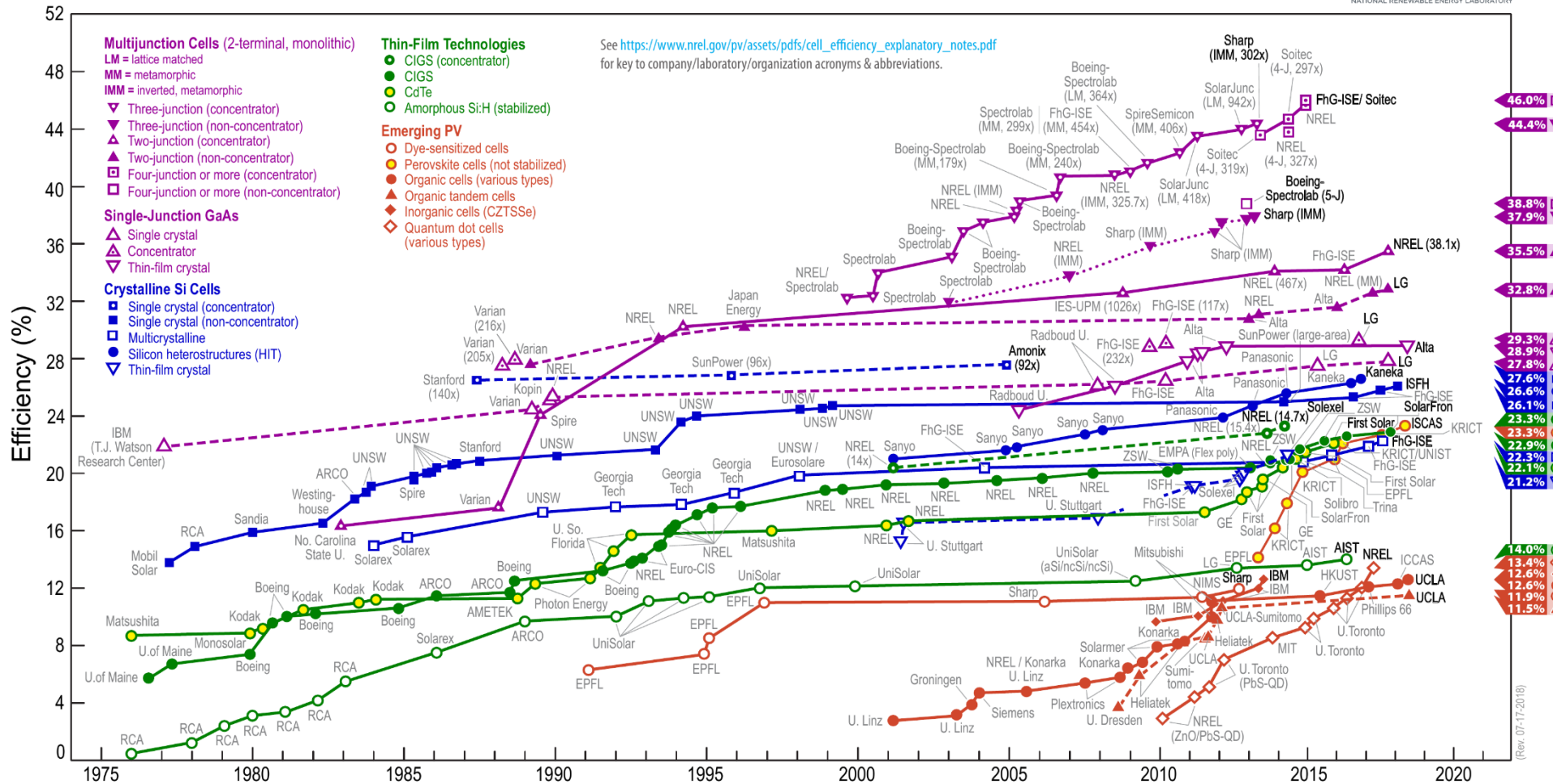
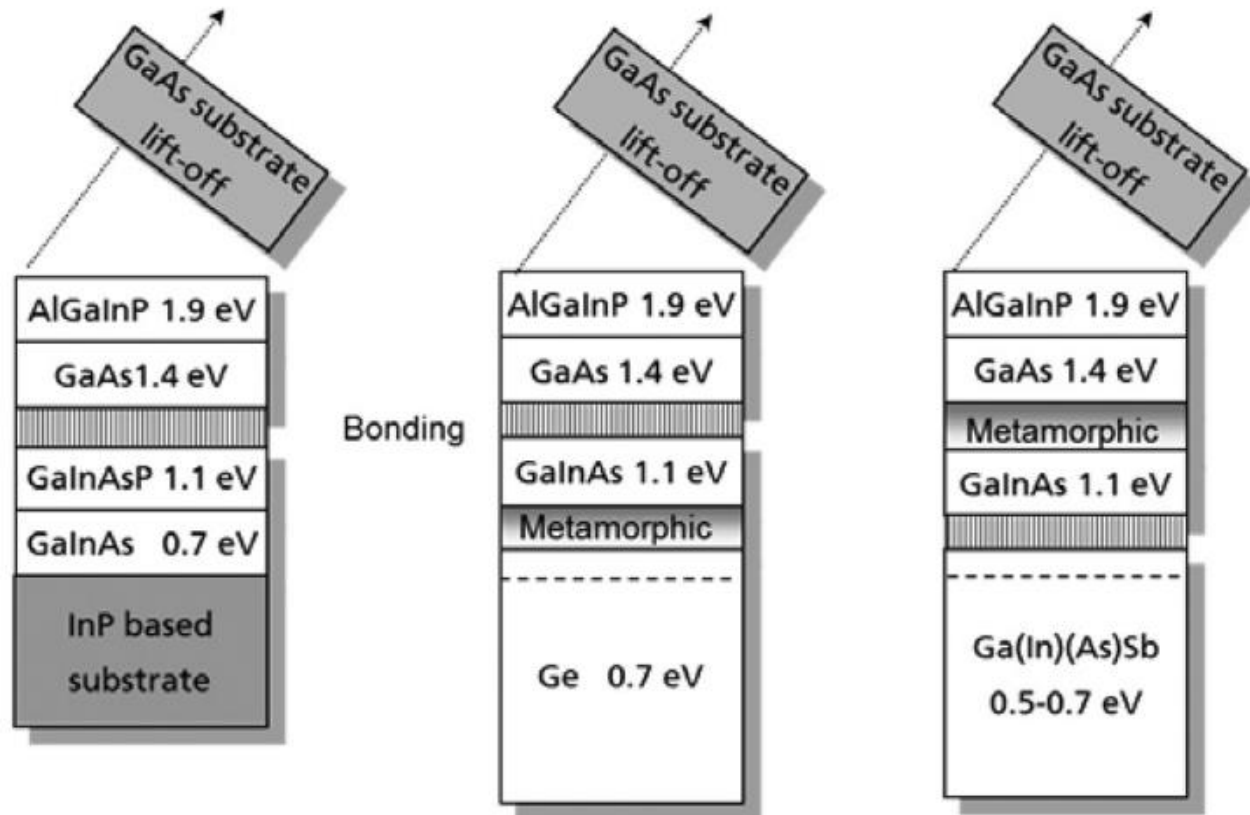


Figure 8.21 (a) Inverted metamorphic three-junction cell as grown. The tunnel-junction interconnects are not shown. The drawing is not to scale. (b) Finished device structure after back-contact deposition, handle mounting, substrate removal, and front-contact deposition

Solar Cell Efficiency records (as in 2018)

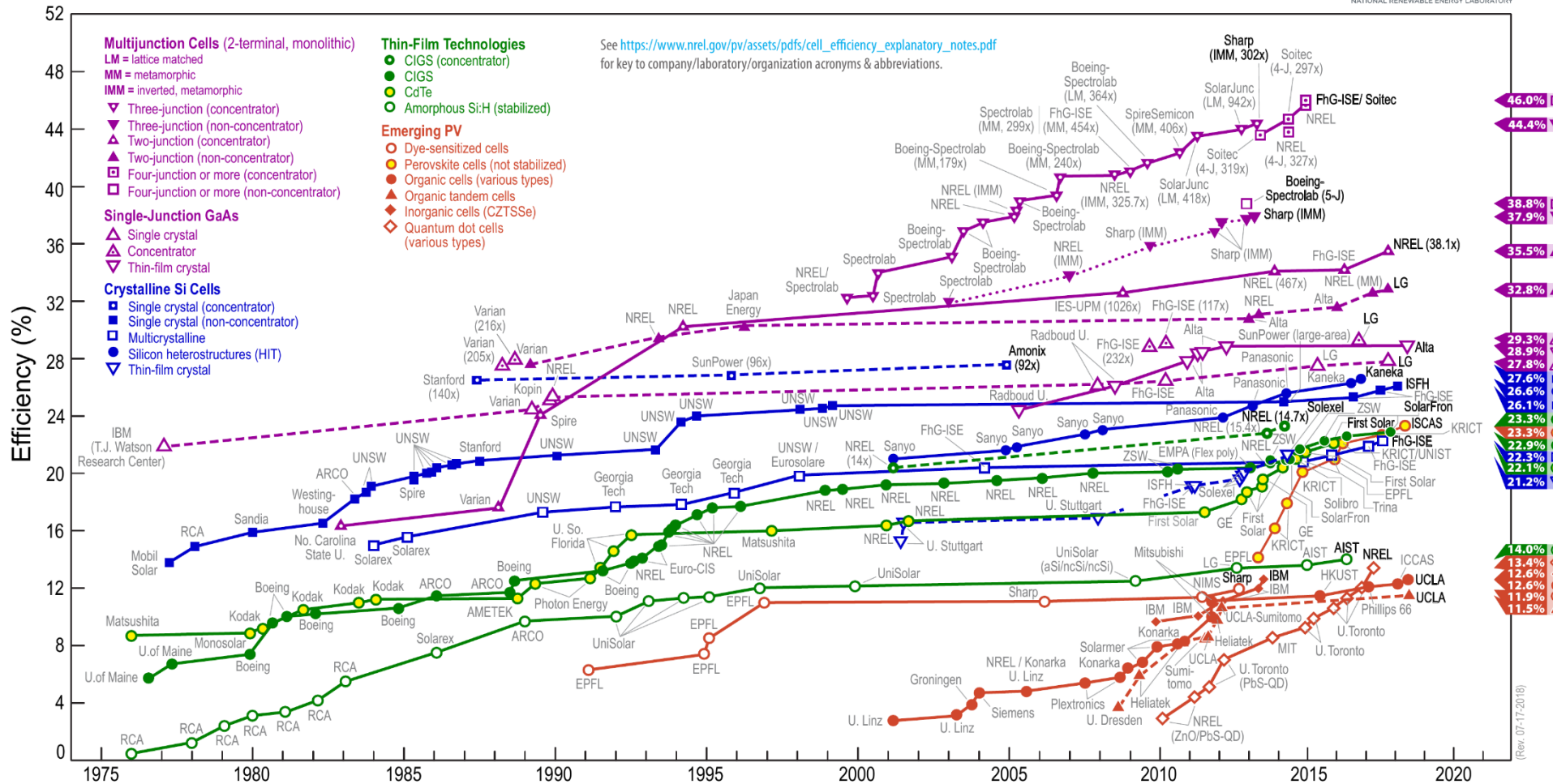
Best Research-Cell Efficiencies



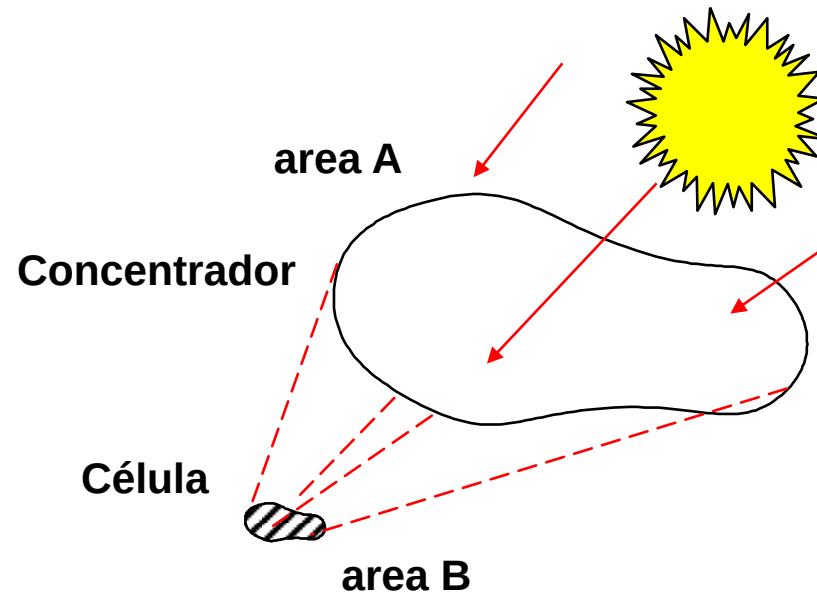


Solar Cell Efficiency records (as in 2018)

Best Research-Cell Efficiencies



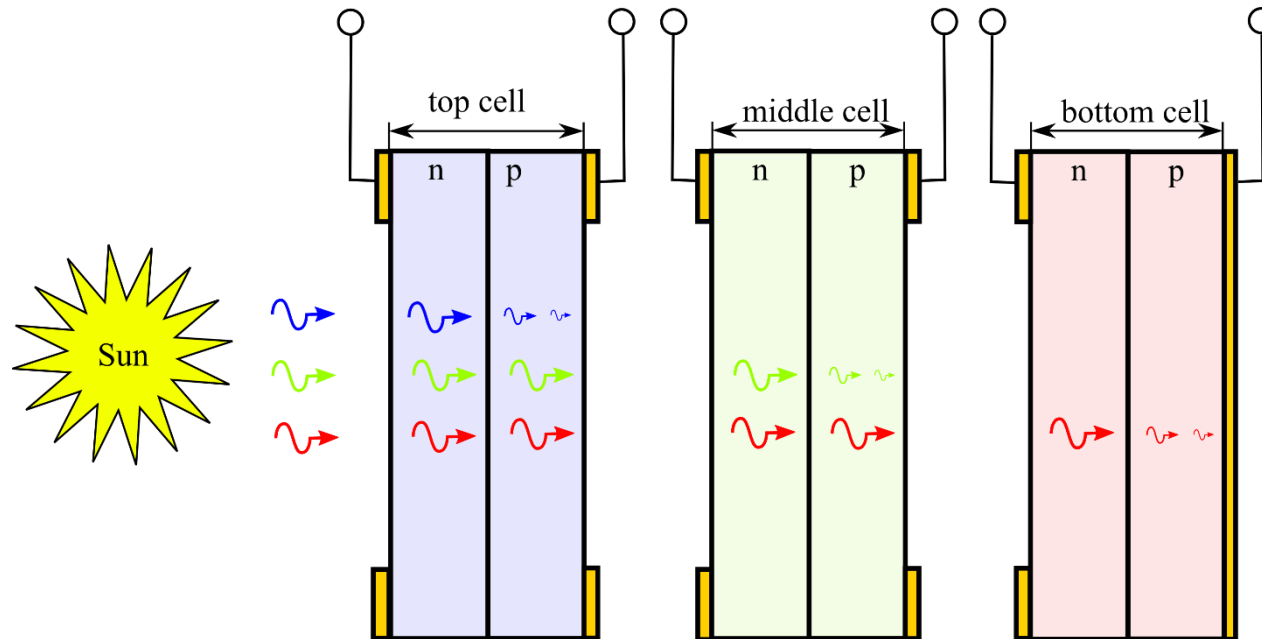
Used in concentration systems

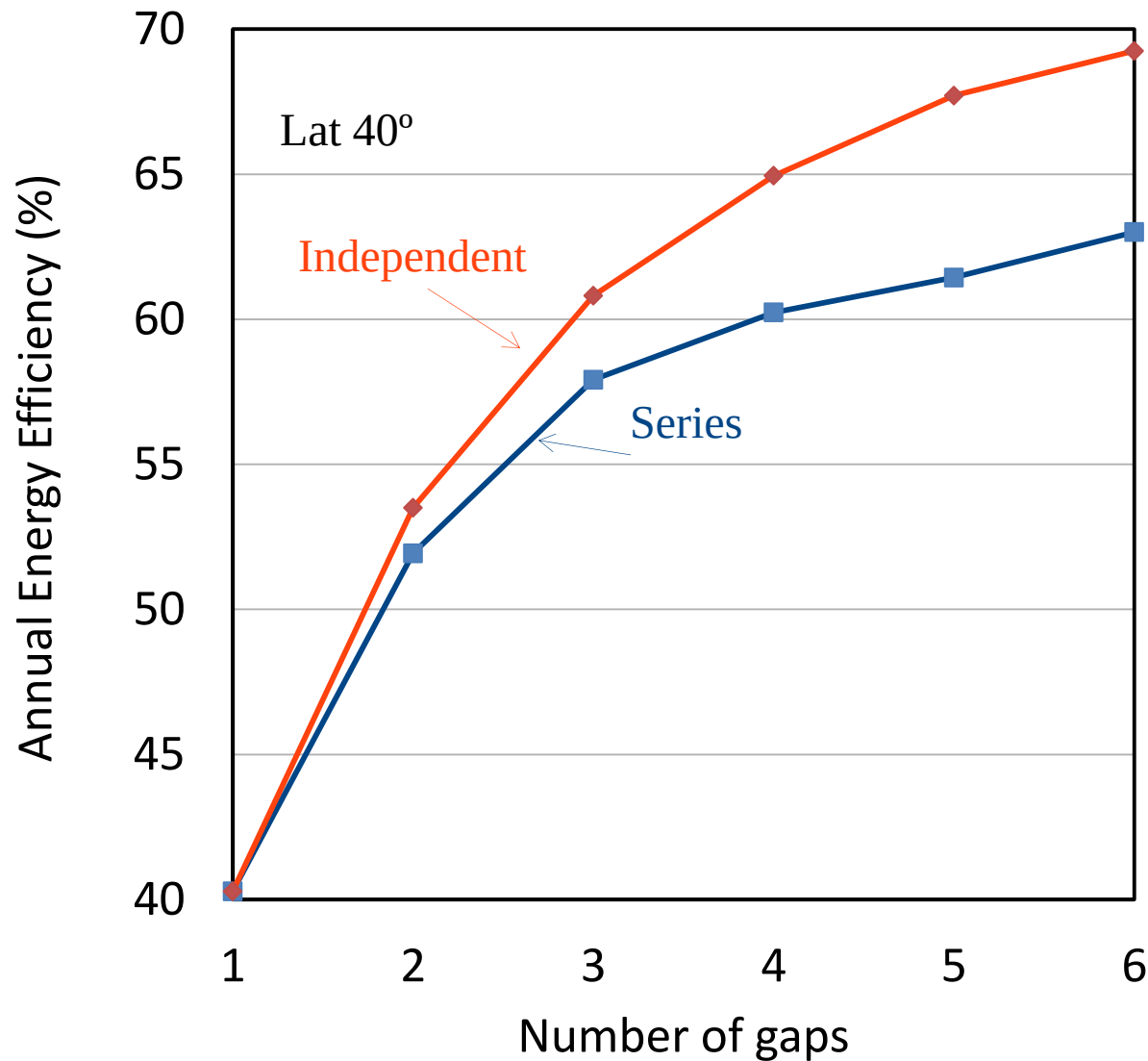


Concentration

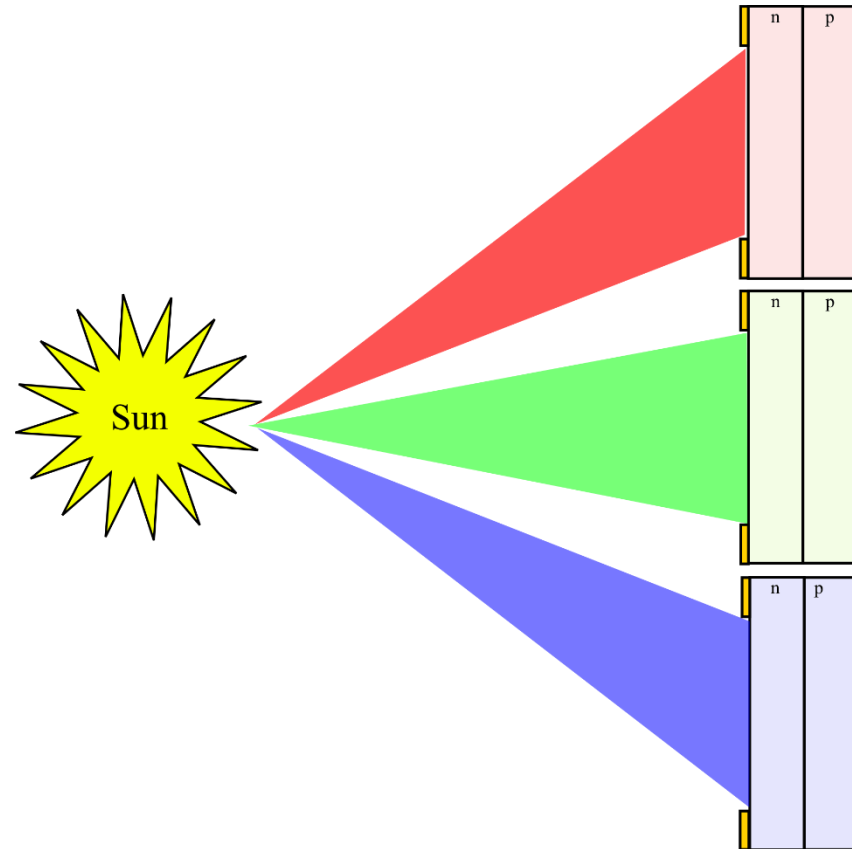


Tandem cells: independent connexion





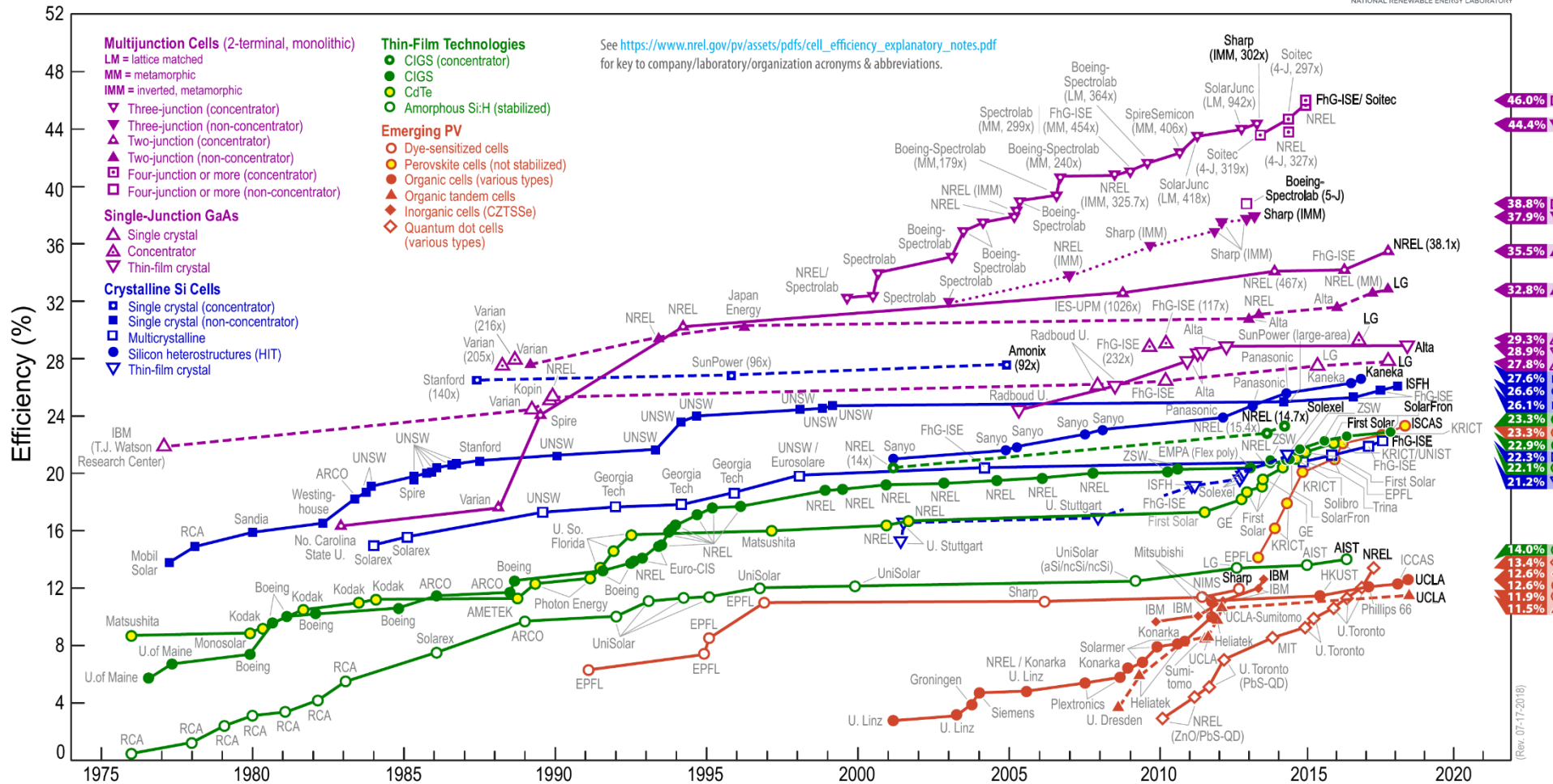
Tandem cells: spectrum splitting



- Fundamentals of photovoltaic Energy conversion
- Conventional (inorganic) solar cells
 - silicon
 - III-Vs (multi-junction solar cells, GaAs, InGaP, InGaAs...)
 - thin films
 - CdTe
 - CIGS
 - (a-Si)

Solar Cell Efficiency records (as in 2018)

Best Research-Cell Efficiencies



- Strong absorption coefficient
- Deposited as polycrystalline materials

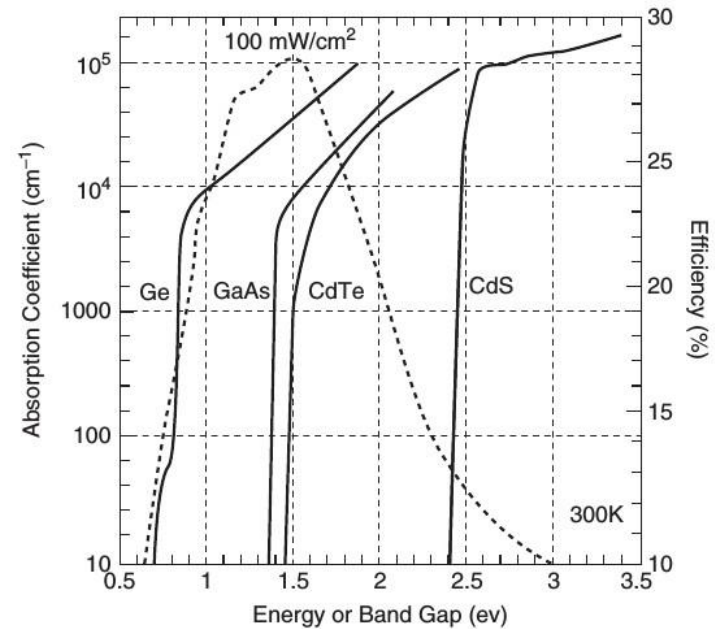


Figure 14.1 Theoretical solar cell efficiency (dotted) for AM1.5 spectral irradiance versus bandgap and absorption coefficient (solid) versus energy. Common absorber materials are highlighted

Brian E. McCandless and James R. Sites, Handbook of photovoltaic Science and Engineering 2 ed. (John Wiley & Sons, Chichester, 2004).

- Strong absorption coefficient
- Deposited as polycrystalline materials on cheap substrates
- Second in the market after silicon

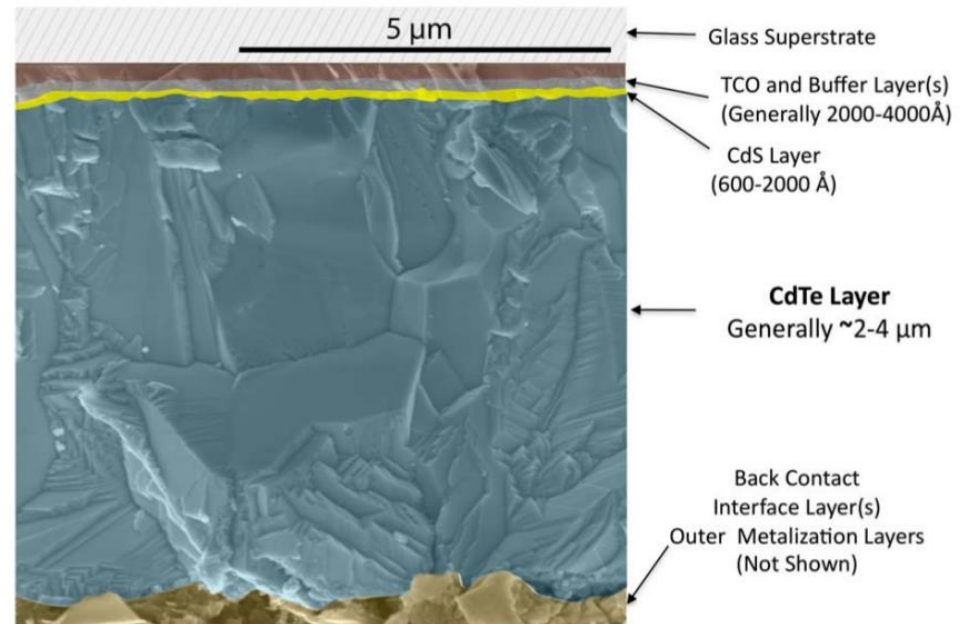


Figure 3.1 SEM micrograph showing cross section of CdS/CdTe superstrate device produced at NREL. Certain layers have been color-enhanced for clarity.

T. Gessert, B. McCandless and
C. Ferekides in A. J. Nozik, G. Conibeer, and M. C. Beard,
*Advanced Concepts in Photovoltaics: Royal Society of
Chemistry, 2014.*

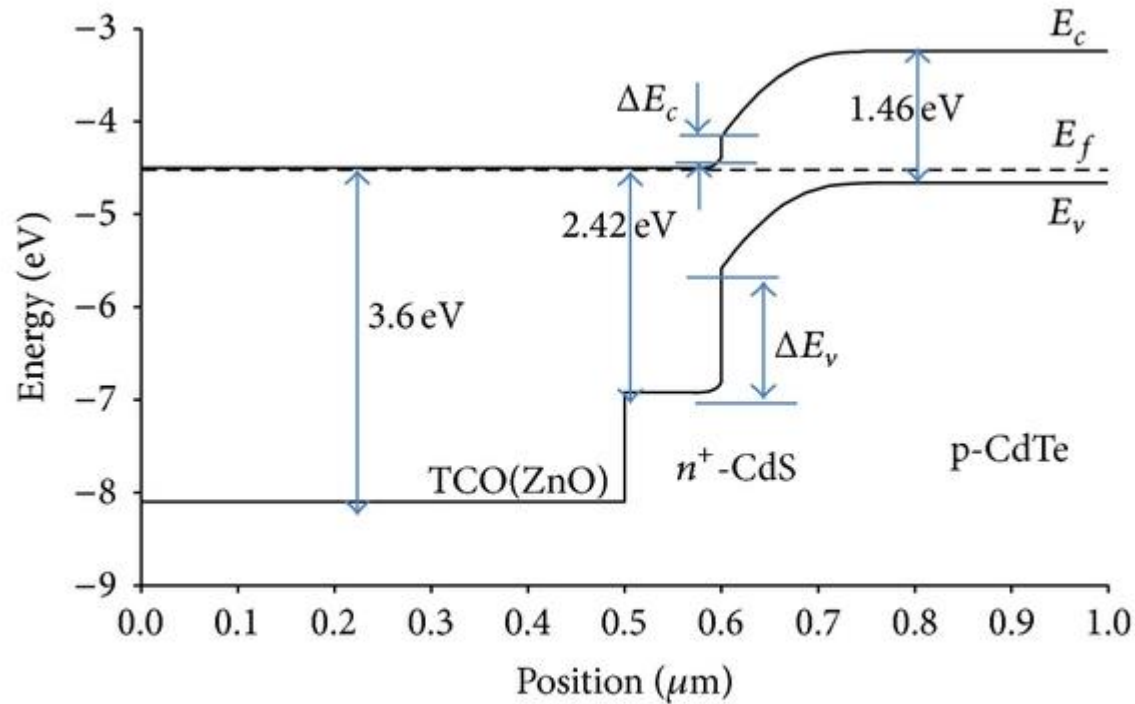
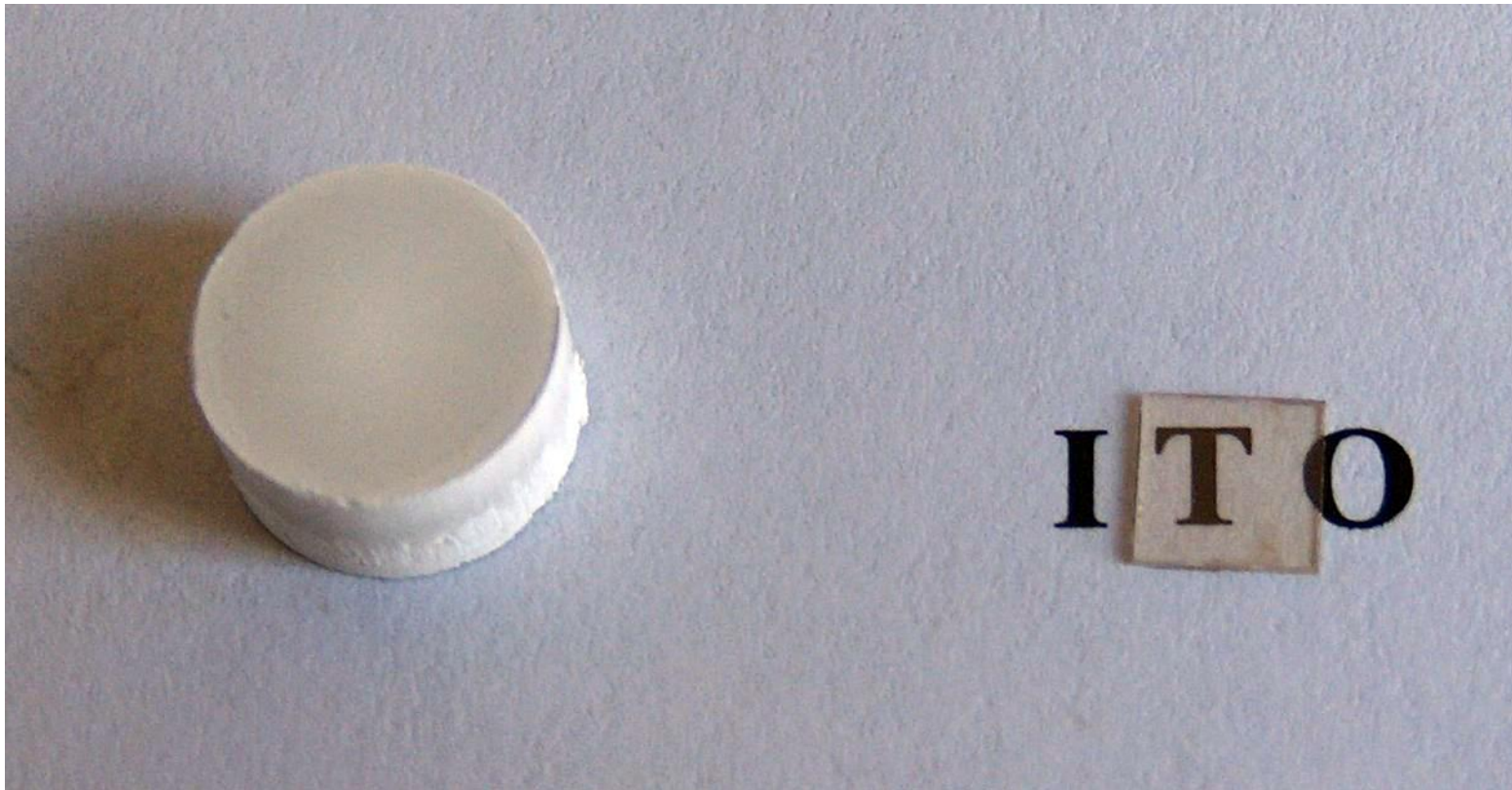


Figure 2: Band diagram of TCO/CdS/CdTe cell without electron reflector layer.

TCO: Transparent conductive oxide

ITO: Indium Tin Oxide, 4 eV



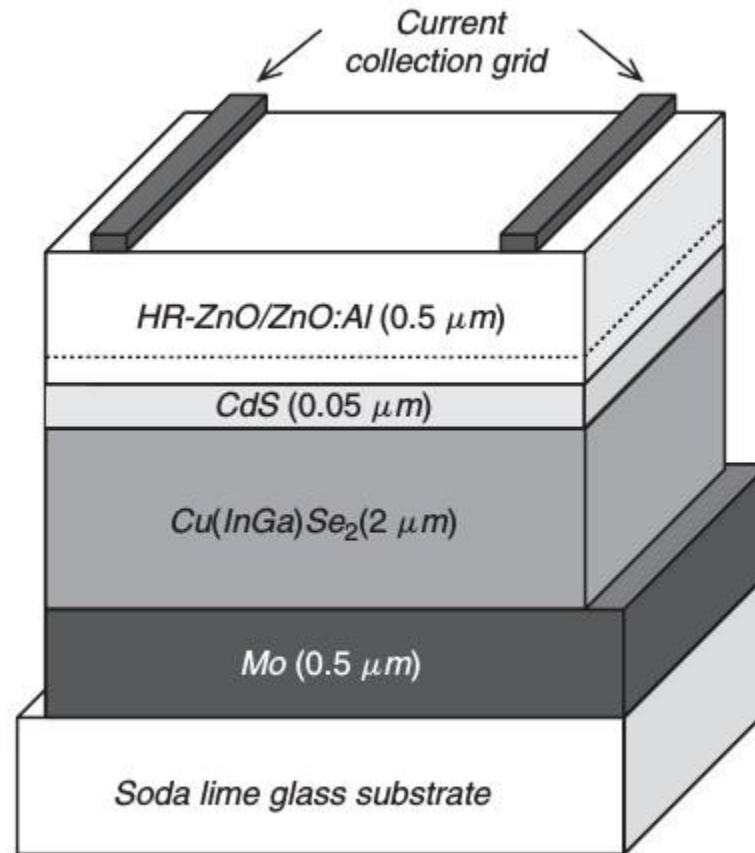


Figure 13.1 Schematic cross-section of a typical Cu(InGa)Se₂ solar cell

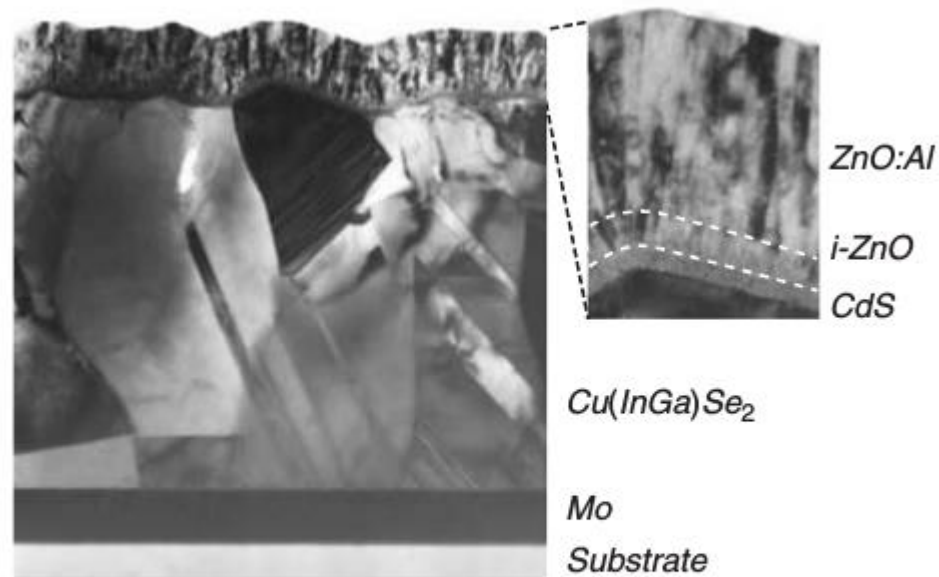


Figure 13.2 TEM cross-section of a Cu(InGa)Se₂ solar cell

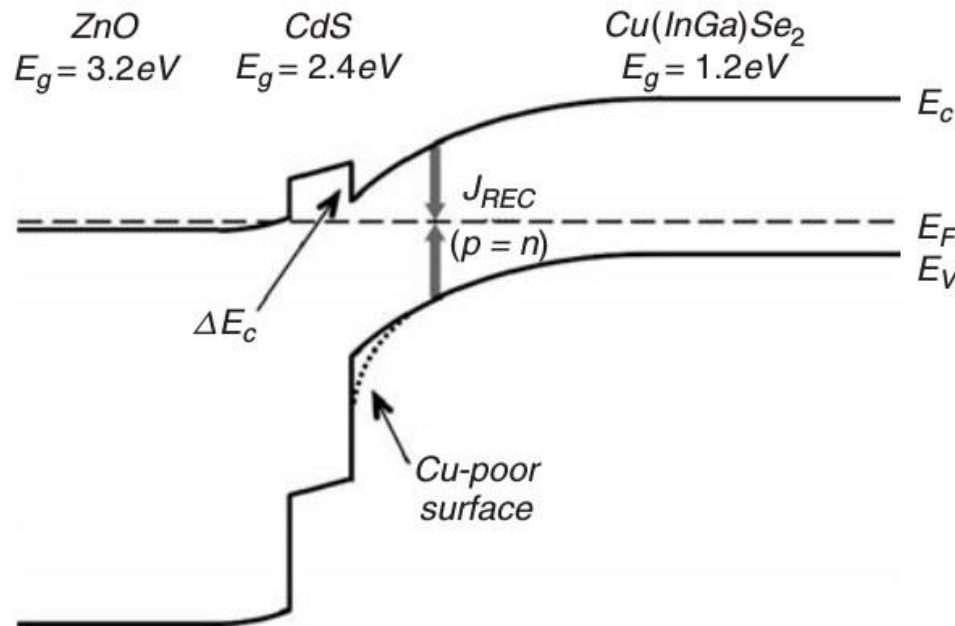


Figure 13.18 Band diagram of a ZnO/CdS/Cu(InGa)Se₂ device at 0 V in the dark. The recombination current J_{REC} is greatest where $p = n$ in the space charge region of the Cu(InGa)Se₂ and not at the interface. A Cu-poor surface layer is shown by a dashed line

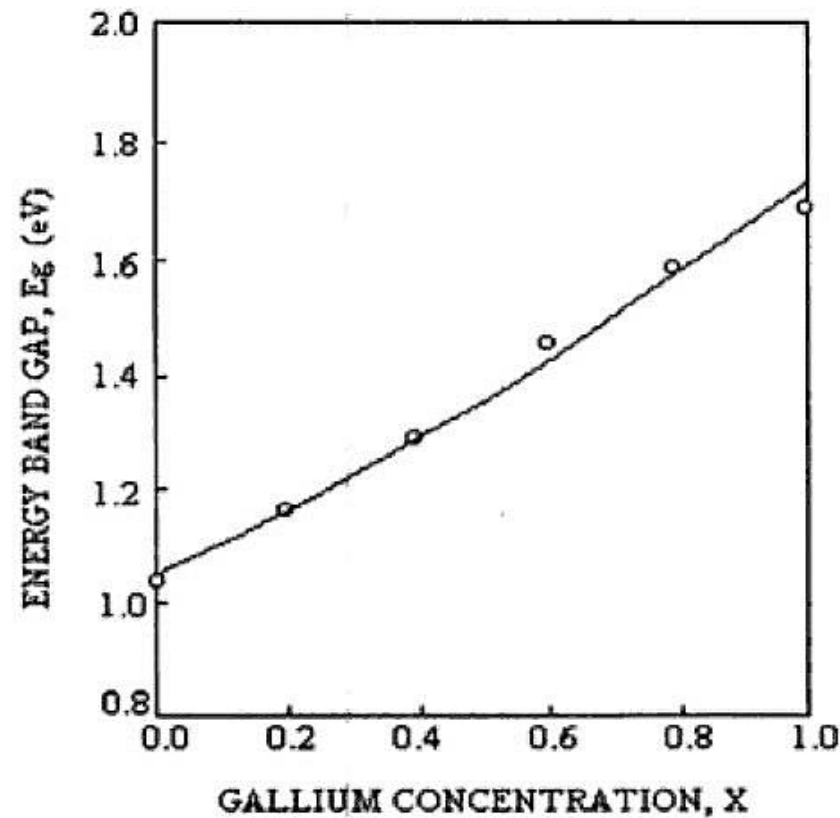
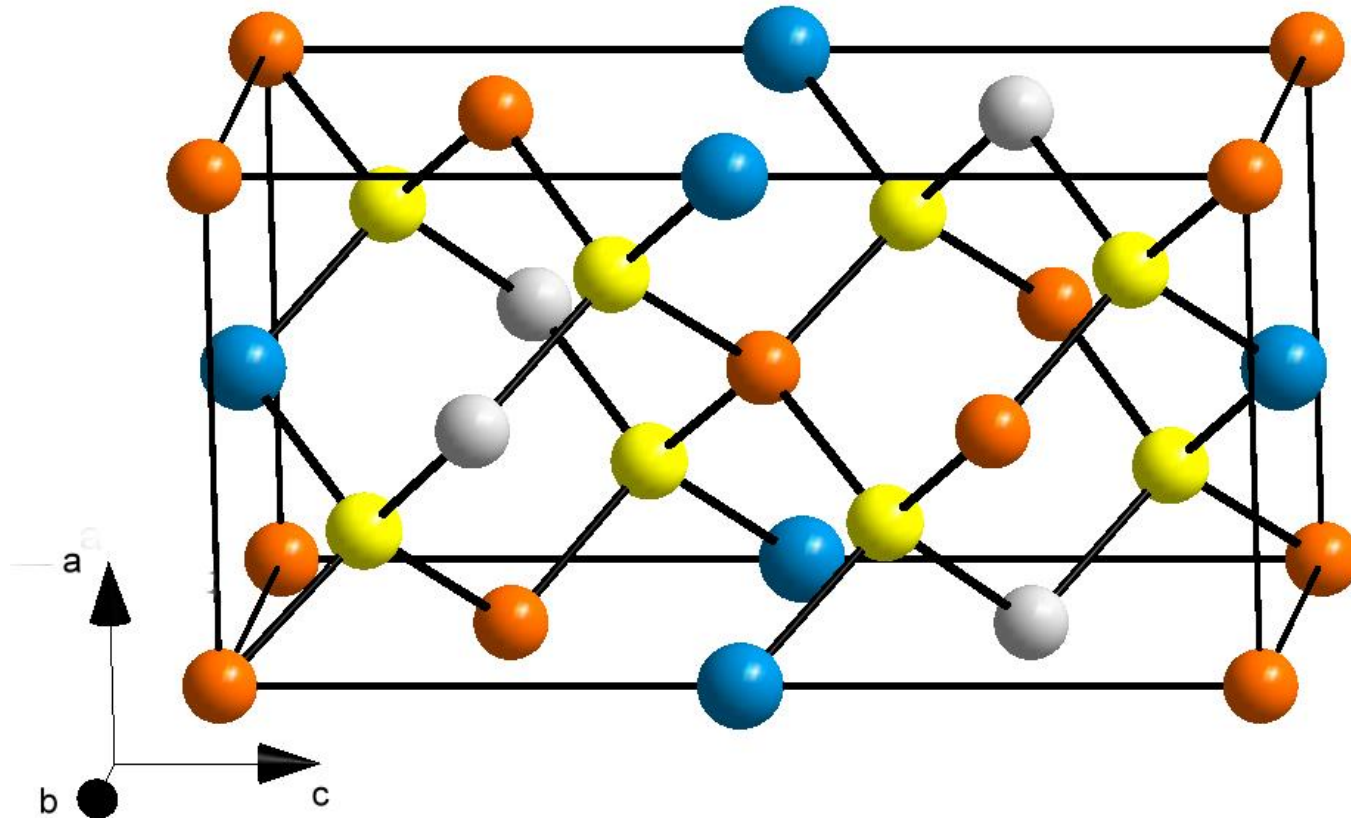


Fig.3. Variation of energy band gap, E_g with gallium concentration in $\text{CuGa}_x\text{In}_{1-x}\text{Se}_2$ films

CTZS: Cupper Zinc Tin Sulphide (kesterites)



Thank you!